

Subject Extraction, Anti-Agreement, and the Unfolding of the CP

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Abstract: This paper documents and analyzes an interaction between subject extraction and the systems that construct the clausal spine in Mandarin, a language of Indonesia. When absolutive subjects undergo WH-movement, they trigger an Anti-Agreement Effect (Ouhalla, 1993; Finer, 1997), forcing the total suppression of τ^0 . This shift is then matched by a series of syntactic effects, from the suppression of complementizers to realignments in the syntax of clausal complementation and an exceptional type of transparency, to suggest that the clauses which launch subject extraction must lack the heads c^0 and τ^0 (Shlonsky, 2014; Pesetsky, 2019). Building on the theory of head-bundling (Rizzi, 1997; Giorgi & Piansesi, 1997), the paper thus develops a derivational system that allows subject extraction to derail the process that unfolds complex heads to construct the clausal spine (Shimada, 2007)—leveraging a particular type of multitasking (Pesetsky & Torrego, 2001; Van Urk & Richards, 2015) to allow A-movement of WH-subjects through regular subject positions to derail the head-movement that typically constructs the TP and CP.

Keywords:

Subject Extraction, Anti-Agreement, Comp-Trace Effects, Hyperraising, Head-Bundling, Reprojection

1 Introduction

The standard theory holds that clauses are built around a three-part functional spine: the heads that make up the extended VP (Larson, 1988; Johnson, 1991; Kratzer, 1996) are dominated by a layer of heads that form the extended TP (Pollock, 1989; Jonas & Bobaljik, 1993; Bobaljik & Thráinsson, 1998), and that layer, in turn, falls beneath a higher series of heads that make up the extended CP (Branigan, 1996; Rizzi, 1997). The frameworks that adopt this view typically assign these layers consistent and distinct syntactic roles (Chomsky, 1995; Platzack, 1999; Grohmann, 2003; Ramchand & Svenonius, 2014): the heads of the extended TP define the syntax of subjecthood and trigger A-movements, while those of the CP establish clausal force and trigger $\bar{A}$ -attraction (Chomsky, 1981, 2008; Rizzi, 1997). But the typical syntax of these domains is routinely disrupted, and the partition between them blurred, in the arena of subject extraction: when WH-subjects are poised to interact with the heads of these two domains, they often skip expected positions in the extended TP (Rizzi, 1982; Brandi & Cordin, 1989; Campos, 1997; McCloskey, 2000), often extracting from the VP (1a), and force the suppression of heads in the extended CP (Perlmutter, 1968; Bresnan, 1972; Chomsky & Lasnik, 1977), yielding comp-trace effects like (1b).

(1) *Puzzles of Subject Extraction*

- a. [_{TP} Who [_{TP} — [_{voiceP} *t* left]]]?
- b. [_{CP} Who do you think [_{TP} (*that) [_{voiceP} *t* left]]]?

The goal of this paper is thus to document and analyze an analogous interaction in the system of subject extraction in Mandarin, an ergative language of Indonesia. This language typically requires the absolutive DP to interact with τ^0 and raise covertly through a series of subject positions in the extended TP, yielding the pattern of absolutive agreement in (2a). But when the absolutive argument is bound for interaction with c^0 —in contexts of

WH-movement, focus-fronting, and relativization—this agreement disappears (2b), yielding an *Anti-Agreement Effect* (Ouhalla 1993; Finer 1997; Shlonsky 2014; Baier 2018).

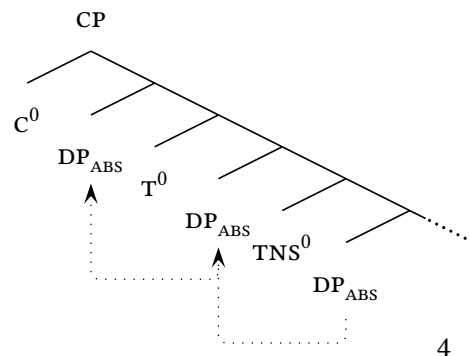
(2) *Mandar: Absolutive Anti-Agreement*

- a. [_{TP} Mu-ita i i'o do setang o]?
 2ERG-see 3ABS you that demon there
 ‘You saw that demon there?’
- b. [_{<TP} Apa mu-ita ___/*i t]]?
 what 2ERG-see 3ABS
 ‘What did you see?’

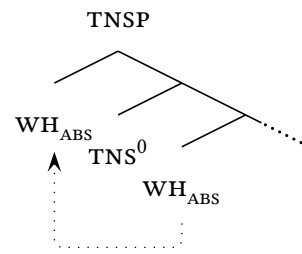
The standard perspectives link Anti-Agreement to two types of syntax: one on which subject extraction skips regular subject positions (Brandi & Cordin, 1989; Schneider-Zioga, 2007; Diercks, 2009; Erlewine, 2016) and another that takes it to pass through regular subject positions and trigger postsyntactic impoverishment (Ouhalla, 1993; Cheng, 2006; Baier, 2018). In Mandarin, however, we will see that Anti-Agreement implicates a syntax that is distinct: the facts of quantifier stranding show that WH-subjects pass through their typical A-positions of the extended TP, but several further effects reveal that it suppresses a band of structure between the extended TP and extended CP (Shlonsky, 2014; Pesetsky, 2019): while regular clauses feature a TP and a CP (3a), these projections go unbuilt whenever WH-phrases raise into a specific A-position along the path the absolutive DP (3b).

(3) *Subject Extraction: Suspended Structure-Building*

a. *Regular Clauses*



b. *Subject Extraction*



To capture this effect in derivational terms, I will argue that certain steps of clausal structure-building must be driven by features that are introduced to the derivation at an early stage and deactivated when WH-phrases are drawn to specific A-positions along the spine. In Mandarin, more specifically, I will argue that the heads τ^0 and c^0 are canonically introduced into the derivation within complex heads that are bundled together with lower material in the extended TP (Rizzi, 1997), hosting strings of category features that will later unfold to yield functional projections along the spine (Giorgi & Pianesi, 1997; Shimada, 2007). In the usual case, the TP and CP are built through a cyclic process of head-movement and reprojection triggered by features of this type (Fanselow, 2004; Georgi & Müller, 2010; Martinović, 2015)—but when WH-subjects are drawn up into lower A-positions by earlier features on these heads, I propose that this process is suspended by a type of multitasking (Pesetsky & Torrego, 2001; Van Urk & Richards, 2015; Newman, 2024): one that parasitically checks the $\bar{A}$ -features associated with the CP when WH-subjects raise to A-positions in the extended TP. The result is an analysis that leverages head-bundling across the TP and CP to derivationally suspend structure-building when subjects are bound for the CP.

The remainder of this paper is structured as follows. Section 2 provides background on Mandarin, reviews the syntax of subjecthood, introduces the systems of WH-movement and Anti-Agreement, and introduces a pattern of WH-quantifier stranding to show that WH-subjects must extract from a regular subject position. Section 3 then shows that subject extraction forces the suppression of c^0 and τ^0 : when it proceeds from embedded clauses, it triggers a comp-trace effect and renders them transparent to various types of extraction that are blocked by τ^0 . Section 4 then introduces the unfolding framework and develops a particular claim about the nature of subject extraction: the process that draws WH-subjects to their final positions in Mandarin must be A-movement to a regular subject position, rather than $\bar{A}$ -movement to SPEC,CP. Section 5 then concludes by drawing a link to a classical puzzle on the relationship between WH-movement and raising in Austronesian.

2 Background

Mandar is an Austronesian language spoken by 400,000 people in the Indonesian province of West Sulawesi. It belongs to the South Sulawesi subfamily of Malayo-Polynesian (Esser, 1938; Mills, 1975; Grimes & Grimes, 1987) and has been the subject of a range of descriptive studies in Indonesian (Pelenkahu *et al.*, 1983; Sikki *et al.*, 1987; Muthalib & Sangi, 1991). This paper describes the standard variety of the language, spoken on the coast between the cities of Polewali and Majene (Grimes & Grimes, 1987). Its results emerge primarily from elicitation with a single speaker, Jupri Talib, with whom I have been working since 2018. This work has been conducted primarily over zoom and done in Mandar, with occasional recourse to Indonesian, and then supplemented with secondary elicitation with two native-speaker linguists, Sitti Sapiyah and Hairuddin, as well as regular interactions with the wider Mandar community through a series of visits to Sulawesi over this time.¹

2.1 Clausal Syntax

The typical Mandar clause has the shape in (4): nominal arguments surface in the *voiceP* in a space that linearly follows the verb and precedes adjuncts to the *voiceP*. In this space, they take the rigid order of S-O-D: the EXTERNAL ARGUMENT (EXT) precedes the INTERNAL ARGUMENT (INT) and the INT precedes the APPLIED ARGUMENT (GOAL) in ditransitives. While these arguments are not marked with morphological case, they trigger agreement: transitive and ditransitive verbs show ergative agreement with the EXT, while every finite clause hosts a second-position clitic that agrees with the absolutive DP (underlined in Section 2.1): the intransitive EXT or INT, the transitive INT, or the ditransitive GOAL (4).

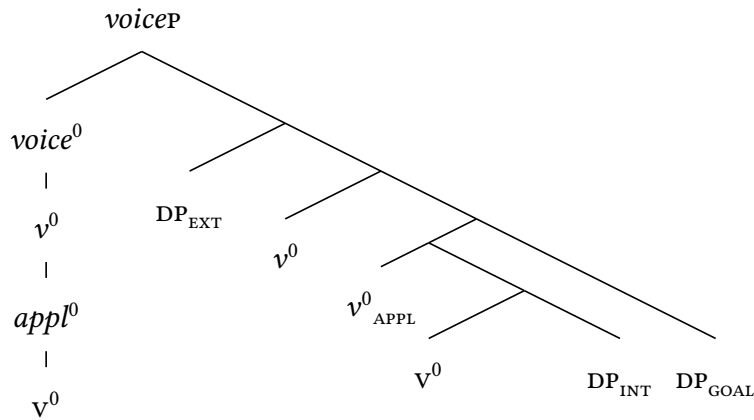
¹GLOSSING: ABS: absolutive, ANTIP: antipassive, APPL: applicative, ERG: ergative, GEN: genitive; PASS: passive; PRT: particle; REL: relativizer; SG: singular. ORTHOGRAPHY: <c, ng, ' >: /tʃ, ŋ, ʔ/.

(4) *The Mandar Clause*

- a. Byasa i [voiceP na-alliang [EXT iKaco'] [INT bunga] [GOAL iCicci']] dio.
 usually 3ABS 3ERG-buy NAME flowers NAME there
 'Kacho' usually buys Chichi' flowers there.'
- b. Byasa o [voiceP u-alliang [EXT pro] [INT bunga] [GOAL pro]] dini.
 usually 2ABS 1ERG-buy 1SG flowers 2SG here
 'I usually buy you flowers here.'

Brodkin 2025a argues that the Mandar *voiceP* shows the syntax in (5): the EXT is introduced in the specifier of v^0 (Collins, 2005; Merchant, 2013), the GOAL is introduced in a rightward specifier of a lower *appl*⁰ (Pylkkänen, 2008), and the verb raises through these heads to *voice*⁰ by a type of word-building head-movement (Harizanov & Gribanova, 2019). This process amalgamates it with affixes in three heads: the ergative prefixes in *voice*⁰, a series of transitivity suffixes in v^0 , and the applicative suffix *-ang* in *appl*⁰. On this view, the order of $v > \text{EXT} > \text{INT} > \text{GOAL}$ emerges from spell-out of the verbal complex in *voice*⁰ and the consistent realization of the EXT, INT, and GOAL in their base positions.

(5) *The Mandar VoiceP*



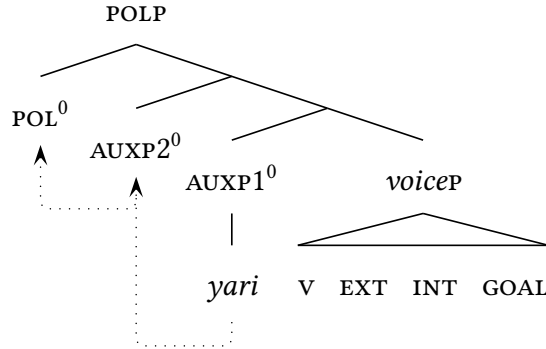
(6) *Heads above the VoiceP*

- a. [_{POLP} **Ndang** i [_{AUXP2} rua [_{AUXP1} yari [_{voiceP} na-baca] dionging]]].
 not 3ABS ever DID 3ERG-read yesterday
 ‘She never DID read it yesterday.’
- b. [_{TNSP} **Allo sattu** i [_{AUXP2} byasa [_{AUXP1} pura [_{voiceP} na-papia] —]]].
 saturday 3ABS usually done 3ERG-make
 ‘By Saturday she’s usually done making it.’

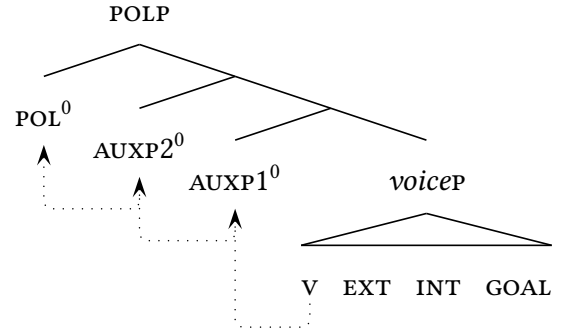
Within this space, there is a second pattern of head-movement that targets the head just beneath TNS⁰, identified here as POLARITY⁰ (abbreviated POL). This process targets the highest auxiliary that overtly appears (7a), and otherwise it draws up the verb (7b).

(7) *Head-Movement to Polarity*

a. *Auxiliaries*



b. *Verb-Raising*



The highest overt element in TNSP, in turn, hosts the agreement enclitics that target the absolutive DP (as across the South Sulawesi subfamily: [Strømme 1994](#); [Friberg 1996](#); [Finer 1999](#)). This pattern converges with many further effects to suggest that absolutive agreement sits in a head above TNS⁰: it appears in all finite clauses, surfaces as a suffix on the irrealis c⁰ *anna*’ and takes a distinct irrealis paradigm in that context (8a), and disappears in reduced clauses like control complements (8b). To capture these effects, [Brodkin 2022](#) proposes that it sits in T⁰ and reaches its surface position through a postsyntactic process that inverts it with the highest element beneath that head (see e.g., [Halpern 1995](#); [Embick & Noyer 2001](#)): an adverb in SPEC,TNSP or an auxiliary or raised verb in POL⁰.

(8) *Absolutive Agreement: High*

- a. Pole **a'** yau marondong [CP **anna'**_{C⁰} -u_{T⁰} mala mu-ita].
 come 1ABS 1SG tomorrow so.that-1ABS.IRREALIS can 2ERG-see
 'I'll come tomorrow so that you might see me.'
- b. Tapi' batabata **a'** [POLP na-ita ___/***a'**_{T⁰} kindo'-u].
 but reluctant 1ABS 3ERG-see 1ABS mom-1GEN
 'But I'm reluctant for my mom to see ____{PRO}'

The high position of absolutive agreement is then matched by a corresponding process of raising that targets the absolutive DP. This movement is typically masked by the preference to spell out arguments in their base positions, yielding the consistent word order of V>EXT>INT>GOAL. But it is consistently reflected in the systems of variable binding and Condition C (Section 3), and we can also see it linearly with a series of strategies that force higher spell-out for the absolutive DP. At the right edge of the intonational phrase, for instance, there is a type of accent that falls on specific clause-final particles and several types of foci (9a). When this accent is placed after a position that should host the absolutive DP, it forces the absolutive DP to be spelled out in the next-highest position on its path. When placed at the right edge of a ditransitive *voice*P, for instance, it prevents the absolutive GOAL from following the INT in the final SPEC,*appl*P. Instead, it must precede that DP (9b), being spelled out in a position we can identify as an inner SPEC,*vp*.

(9) *Clash Effects: Absolutives Raise to SPEC,VP*

- a. Byasa i na-alli-ang [vp iKaco' [applP bunga]] **diò!**
 usually 3ABS 3ERG-buy-APPL NAME flower THERE
 'Kacho' usually buys her flowers THERE!'
- b. Byasa i na-alli-ang [vp iKaco' iCicci' [applP bunga ___]] **diò!**
 usually 3ABS 3ERG-buy-APPL NAME NAME flower THERE
 'Kacho' usually buys Chichi' flowers THERE!'

When we force the same type of accent to follow this higher position, in turn, we can

see a second step: the absolutive GOAL must be spelled out in *SPEC,voiceP*. As a result, it surfaces just above the verb in the clauses where raising to POL^0 is blocked by the presence of auxiliaries (10a). In the clauses that feature verb-raising, in turn, the verb precedes this second position for the absolutive DP (10b).

(10) *Clash Effects: Absolutives Raise to SPEC,voiceP*

- a. Byasa i _[voiceP] iCicci' na-alli-ang _[vP] iKaco' — *pro*_{INT} —]] **diò!**
 usually 3ABS NAME 3ERG-buy-APPL NAME there
 'Kacho' usually buys them for Chichi' THERE!'
- b. _[POLP] Na-alli-ang i _[voiceP] iCicci' _[vP] iKaco' — _[applP] —]]]] **àe!**
 3ERG-buy-APPL 3ABS NAME NAME PRT
 'Kacho' bought some for Chichi', I'm telling you!'

When we force the same type of accent to fall immediately after *SPEC,voiceP*, at last, we see the step in (11): the absolutive GOAL is forced to surface above POL^0 and beneath the temporal adverbs in *SPEC,TNSP*. [Brodkin 2025b](#) identifies this third position as *SPEC,ASPP*, and we will return to the syntax of this projection and the clitics that head it in Section 3.

(11) *Clash Effects: Third Position for the Absolutive DP*

- a. _[ASPP] iCicci' _[POLP] na-alli-ang i _[voiceP] — *pro*_{EXT} *pro*_{INT}]]]] **àe!**
 NAME 3ERG-buy-APPL 3ABS PRT
 'He bought some for Chichi', I'm telling you!'
- b. _[TNSP] Dionging i _[ASPP] iCicci' _[POLP] na-alli-ang _[voiceP] —]]]] **àe!**
 yesterday 3ABS NAME 3ERG-buy-APPL PRT
 'He bought some for Chichi' yesterday, I'm telling you!'

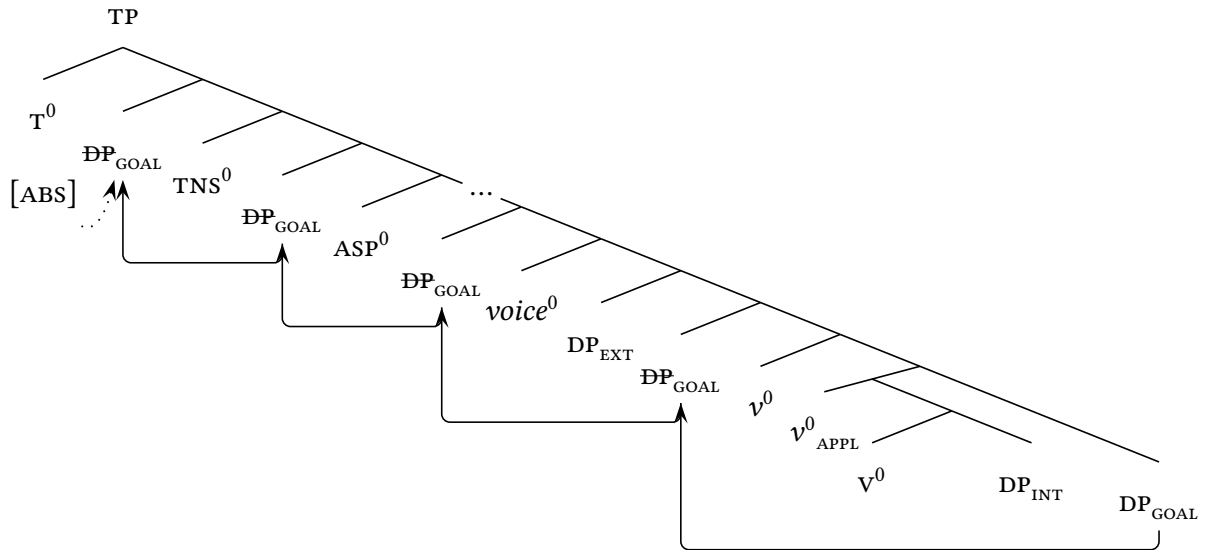
We can then see a final step of movement with a separate manipulation. Mandarin has a series of prenominal quantifiers that derail the usual preference for lowest-copy spell-out and force their complements to be realized in the highest positions they pass through. When quantifiers of this type select the absolutive DP, it must be spelled out above the fronted adverbs in *SPEC,TNSP* (12a), rather than in any position lower in the clause (12b).

(12) *Quantified Absolutives: Fourth Position for the Absolutive DP*

- a. $[_{TNS} \text{Ing. tau} \text{ diwongi } i \text{ } [_{ASPP} \text{ — } [_{POLP} \text{ na-be-ngang ing. ande }]]]$.
 everyone last night 3ABS 3ERG-give-APPL all food
 ‘Last night he gave everyone all the food.’
- b. $*[_{TNS} \text{ — } \text{Diwongi } i \text{ } [_{ASPP} \text{ ing. tau } [_{POLP} \text{ na-be-ngang ing. ande }]]]$.
 last night 3ABS everyone 3ERG-give-APPL all food
 INTENDED ‘Last night he gave everyone all the food.’

These facts suggest a clausal syntax familiar from the wide class of ergative languages that require the absolutive DP to receive the highest source of structural Case (Coon *et al.*, 2014): one that forces this DP to raise through a series of A-positions on a path up to T^0 . The path of raising in a ditransitive clause is sketched in (13): the absolutive DP undergoes a short step of object shift to an inner SPEC, νP , then raises to the low subject position of SPEC, $\text{voice}P$, then moves further to SPEC, ASPP before finally raising to SPEC, TNSP .

(13) *The Syntax of Subjecthood*



From this position, we can draw a bridge to the morphosyntactic alternations that make up the Austronesian system of *voice* (Guilfoyle *et al.* 1992; Aldridge 2004; Nomoto 2013; Legate 2014; Erlewine 2018; Erlewine & Sommerlot To Appear; Nie 2020). In the

traditional terminology, the ditransitive verbs that we have studied above are ones that surface in the LOCATIVE VOICE: a construction that requires absolutive agreement to fall on the GOAL. These verbs contrast with the regular transitives of the language, which reflect the traditional PATIENT VOICE: a construction that requires absolutive agreement to target the INT (14a). In the presence of these verbs, the syntax of subjecthood carries up the INT: when quantified, for instance, it begins to raise overtly to SPEC,TNSP (14b).

(14) *Absolutive Movement: Targets the Transitive INT*

- a. Diwongi a' na-telpong iKaco' *pro*.
 last night 1ABS 3ERG-call NAME 1SG
 'Last night Kacho' called me.'
- b. [_{TNSP} **Ing. tau** **diwongi** i [_{ASPP} — [_{POLP} na-telpong iKaco']]].
 everyone last night 3ABS 3ERG-call NAME
 'Last night Kacho' called everyone.'

These constructions also contrast with an antipassive which corresponding to the traditional AGENT VOICE (for discussion, see Guilfoyle *et al.* 1992; Aldridge 2004; Paul & Travis 2006). Morphologically, this construction is formed by replacing the ergative prefixes on the verb with a series of invariant affixes like *mang-* (15a). The clauses built around these verbs then differ from those built around transitives and ditransitives in two respects: they require the INT and GOAL to be indefinite and allow them to be implicit and they require the systems of absolutive agreement and raising to target the EXT (15b).

(15) *Absolutive Movement: Targets the Antipassive EXT*

- a. Diwongi a' man-dundu *pro* ballo']]].
 last night 1ABS ANTIP-drink 1SG palm wine
 'Last night I was drinking palm wine '
- b. [_{TNSP} **Ing. tau** **diwongi** i [_{ASPP} — [_{POLP} man-dundu ballo']]].
 everyone last night 3ABS ANTIP-drink palm wine
 'Last night everyone was drinking palm wine '

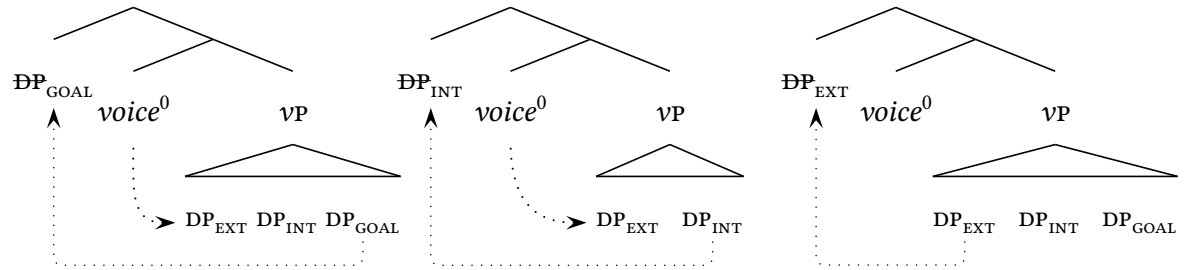
The result is a syntactic alternation within the *voiceP*, bound up with the morphological exponents of *appl*⁰, *v*⁰, and *voice*⁰, that determines the identity of the absolutive DP. When the verb is transitive or ditransitive, *voice*⁰ triggers ergative agreement with the EXT and then attracts the next-highest argument to SPEC,*voiceP*: the GOAL when *appl*⁰ is present (16a) and the INT when it is not (16b). When the verb is antipassive, in contrast, *voice*⁰ triggers no such agreement of this type—and instead attracts the EXT to SPEC,*voiceP* (16c). Each of these steps then feeds the higher sequence of movements through SPEC,ASPP and SPEC,POLP—raising all types of absolutive DP to the position where they will receive Case from T⁰ (and for similar views across South Sulawesi: [Finer 1999](#); [Béjar 1999](#); across Western Indonesia, [Nomoto 2013](#); [Erlewine 2018](#); [Erlewine & Sommerlot To Appear](#)).

(16) *The Alternations in Voice*

a. *Ditransitives*

b. *Transitives*

c. *Antipassives*



2.2 The Anti-Agreement Effect

From this position, we can turn to the syntax of subject extraction: the process that targets WH-phrases that should pass through the higher positions of the absolutive DP. The first step in this connection is to establish the rough syntax of WH-movement, which stands out in regional perspective in several respects. Like Indonesian and Malay ([Cole & Hermon 1998, 2005](#); [Cole et al. 2008](#)), Mandar allows argument WH-questions to be formed around pseudoclefts in the typical Western Austronesian way ([Potsdam, 2009](#)): nominal WH-phrases can take a clause-initial position, function as nonverbal predicates

of copular clauses, and be followed by headless relative subjects. Example (17a) shows a WH-pseudocleft of this type: it requires the WH-predicate to host absolutive agreement and forces the following subject to be headed by an overt demonstrative (17a), on a par with the pseudoclefts built around regular predicate nominals (17b).

(17) *Wh-Pseudoclefts: The Regional Pattern*

- a. $[_{PREDP} \text{ Apa } \mathbf{ri} \quad [_{DP} \mathbf{do} \quad [_{REL.CLAUSE} \text{ mu-alli } __] \text{ o }]]?$
 what JUST.3ABS that 2ERG-buy there
 ‘What’s that thing you bought there?’
- b. $[_{PREDP} \text{ Ande } \mathbf{ri} \quad [_{DP} \mathbf{de} \quad [_{REL.CLAUSE} \text{ u-alli } __] \text{ e }]]$.
 food JUST.3ABS this 1ERG-buy here
 ‘This thing I bought here is food.’

Despite this fact, Mandar prohibits WH-in-situ: a property common to the languages of South Sulawesi (Friberg 1996; Finer 1997; Jukes 2006; Laskowske 2016) and one that sets them apart from the languages of Western Indonesia: (Saddy, 1991; Cole & Hermon, 1998; Cole *et al.*, 2005; Davies, 2000, 2003; Davies & Kurniawan, 2013; Sato, 2008, 2024; Crouch, 2009; Jeoung, 2018; Kim *et al.*, 2019; Middleton, 2024). The following examples illustrate this fact with the WH-adverb *pirang* “when:” this elements cannot take the typical *voiceP*-final position of regular temporal adjuncts and must move past SPEC,TNSP to a left-peripheral position (18). For now, I will refer to this landing site as SPEC,CP.

(18) *Wh-Movement*

- a. $[_{CP} \mathbf{Pirang} \text{ o } \quad [_{TNSP} \text{ dionging } \quad [_{POLP} \text{ na-pelambi'i iKaco' } __]]]?$
 when 2ABS yesterday 3ERG-visit NAME
 ‘When did Kacho’ visit you yesterday?’
- b. $[_{TNSP} \text{ *Dionging o } \quad [_{POLP} \text{ na-pelambi'i } \quad [_{voiceP} \text{ iKaco' }] \quad \mathbf{pirang}]]?$
 yesterday 2ABS 3ERG-visit NAME when
 INTENDED: ‘When did Kacho’ visit you yesterday?’

The same requirement forces WH-subjects to raise in a seemingly parallel way. When WH-phrases are generated in the position of the absolutive DP, they move to a similar position in the left periphery (19a). Unlike the predicates of WH-pseudoclefts, however, these WH-subjects are never followed by absolutive agreement—and the constituents that follow are never headed by demonstratives or relativizers like *anu*, the element which heads free relatives in a manner analogous to the Indonesian *yang* (Saddy, 1991).

(19) *Absolutive Wh-Movement*

- a. $[_{CP} \text{ Apa } [_{TN\text{SP}} \text{ dionging } __ / (*i) \text{ mu-alli } __]] ?$
 what yesterday 3ABS 2ERG-buy
 ‘What did you buy yesterday?’
- b. $[_{CP} \text{ Innai } [_{POLP} \text{ mu-be-ngang } __ / (*i) \text{ anu u-alli }] __] ?$
 who 2ERG-give-APPL 3ABS REL 1ERG-buy
 ‘Who did you give what I bought?’

In the clauses that launch subject WH-movement, absolutive agreement disappears. It cannot surface after the initial WH-phrase (20a) and it cannot appear in any lower position, such as after the highest non-subject in the TN_{SP}—establishing an asymmetry with the clauses where the initial element is a regular absolutive DP (11)-(12) (cf. Friberg 1996, p145 on the related language Konjo). This pattern, moreover, is not linked to the shape of the WH-phrase itself: agreement must be suppressed when raised WH-subjects are simple WH-words like *apa* “what” and also when they are D-linked, like *x inna* “which x” (20b).

(20) *The Loss of Absolutive Agreement: Unlinked to D-Linking*

- a. $[_{CP} \text{ Sa'apa buku } __ / (*i) \text{ mu-baca } __] ?$
 how many book 3ABS 2ERG-read
 ‘How many books did you read?’
- b. $[_{CP} \text{ Buku inna } __ / (*i) \text{ mu-baca } __ / (*i) \text{ } __] ?$
 book which 3ABS 2ERG-read 3ABS
 ‘Which book did you read?’

These facts suggest that we are dealing with a particular type of *Anti-Agreement Effect* (Brandi & Cordin, 1989; Ouhalla, 1993): (absolutive) subject agreement is derailed when the (absolutive) WH-subject raises to the left periphery. This pattern is mirrored by a series of famous effects in nominative-accusative systems, including many Northern Italian languages (Brandi & Cordin, 1989), Celtic (Borsley & Stephens, 1989; Borsley & Kathol, 2000; Borsley *et al.*, 2007), Berber (Ouhalla, 1993, 2005; Ouali, 2006, 2008; Shlonsky, 2014), and Bantu (Schneider-Zioga, 1995, 2007; Cheng, 2006; Henderson, 2006, 2013; Diercks, 2009). But it also has clear analogues in many ergative-absolutive systems that seem to show the same type of High Absolutive Syntax: for instance, Karitiâna (Tupian: Storto 1999) and many languages of Sulawesi (see especially Mead 1998; Kaufman 2008). The most important effect of this class, for our purposes, is the cognate pattern in Selayarese (Finer, 1994, 1997, 1998), a language that shows High Absolutive Syntax (Finer, 1999; Béjar, 1999) and an identical Anti-Agreement Effect: when absolutive subjects undergo extraction, they fail to trigger the usual pattern of third-person absolutive agreement in τ^0 (21).

(21) *Absolutive Anti-Agreement: Selayarese*

- a. [_{TP} La-taro' i doe'=injo iBaso' ri lamari].
 3ERG-put 3ABS money=the NAME in cupboard
 'Baso' put the money in a cupboard.' Finer 1997, ex8a
- b. [_{CP} Apa [_{TP} la-taro' ____/*i ____ iBaso' ri lamari]]?
 what 3ERG-put 3ABS NAME in cupboard
 'What did Baso' put in a cupboard.' Finer 1997, ex9a

The initial profile of this Anti-Agreement effect suggests that it is linked to the network of syntactic disruptions that emerge under highest-argument extraction, rather than the range of secondary factors that might disrupt ϕ -agreement with goals of the feature [WH]. As it is unlinked to D-linking, for instance, it must be distinct from the parallel effects that emerge in Trentino and Fiorentino (Brandi & Cordin, 1989), which disappear when WH-

subjects are D-linked (Suñer 1992; Georgi & Hein 2024; though see also Erlewine 2020). In the same vein, it also emerges across the systems of focus-fronting and relativization—suggesting that it is distinct from the disruptions to ϕ -agreement that are triggered by WH-phrases without WH-movement in languages like Abaza (O’Herin 1995, *pace* Baier 2018’s analysis of Selayarese). The shape of subject relativization in Mandar is shown in example (22a): the relativized DP surfaces immediately before the relative clause, invariably without an intervening c^0 (the relativizer *anu* appears only in free relatives, unlike the Indonesian *yang*), and it systematically fails to trigger agreement on the embedded τ^0 . The facts are the same when absolutive DPs undergo focus-fronting to an analogous landing site in the left periphery: in that context, too, they fail to agree with τ^0 (22b).

(22) *The Loss of Absolutive Agreement: Linked to Subject Extraction*

- a. Na-pittama i [_{DP} do setang [_{REL.CLAUSE} u-ita ___/(*i)] o].
 3ERG-seize 3ABS that demon 1ERG-see 3ABS there
 ‘That demon that I saw there seized him.’
- b. [_{CP} **IKACO**’ [_{POLP} na-pittama ___/(*i) [_{DP} do setang o] ___]].
 NAME 3ERG-seize 3ABS that demon there
 ‘That demon seized KACHO’.

To capture this effect, then, we might imagine that it emerges from the same syntax that Brandi & Cordin 1989 propose to underlie its analogues in Trentino and Fiorentino: one on which WH-subjects extract from within the VP (23a), along the lines proposed by Rizzi 1982 (and see also Kenstowicz 1984; Burzio 1986; Campos 1997), rather than passing through the usual subject positions of SPEC,ASPP and SPEC,TNSP (23b).

(23) *Hypothesis One: The Skipping Derivation*

- a. SKIPPING: [_{CP} WH_{SUBJ} ... [_{TNSP} EXPL ... [_{ASPP} ____ ... [_{voiceP} ____ ...]]]
- b. ✗RAISING: [_{CP} ____ ... [_{TNSP} ____ ... [_{ASPP} ____ ... [_{voiceP} ____ ...]]]

2.3 Tracing the Path of Subject Extraction

To test this hypothesis, we can turn to a convenient pattern of WH-quantifier float. Mandarin has a quantifier *pira* that roughly corresponds to the English *all* and often follows WH-phrases in both fragments and the high position we have identified as SPEC,CP (24). The introduction of this element implies that the answer to the WH-phrase must be a plurality and demands an exhaustive “mention-all” answer (Groenendijk & Stokhof 1984; Beck & Rullmann 1999; Schulz & Van Rooij 2006), like the Indonesian collocation WH-*saja*.

(24) The Wh-Quantifier Pira

- a. [_{QP} Apa **pira**]?
 what all
 ‘What all?’
- b. [_{CP} Innai **pira** [_{TNSP} dionging [_{POLP} mu-carita-ngang rasia ____]]]?
 who all yesterday 2ERG-tell-APPL secret
 ‘Who all did you tell the secret to yesterday?’

Unlike the English *all*, however, *pira* only ever appears in the presence of *apa* ‘what,’ *innai* ‘who,’ and *inna* ‘which.’ This quantifier cannot combine with other WH-phrases or with regular DPs, and as a result, the collocation *all*-x must be expressed with the separate quantifier *inggannanna* (25a). The adverbial uses of the English *all*, in turn, are exclusively expressed by the use of *inggannanna* and the distributive adverb *sangnging* (25b).

(25) The Quantifier Pira: No Independent Uses

- a. **Inggannanna** [tau / sola-u / tallu-m-bua] (*tau pira...)
 all person friend-1GEN three-LINKER-fruits
 ‘All people, all my friends, all three fruits’
- b. **Inggannanna** monge’ a, yari **sangnging** di-ala i tugas-u.
 all sick 1ABS, so DISTRIBUTIVE PASS-take 3ABS task-1GEN
 ‘I’m all sick, so my tasks were all taken.’

I take these facts to show that *pira* must be a quantificational head in the extended NP that lexically selects the three WH-phrases above (on lexical selection: Merchant 2019). On this view, it typically forms a surface constituent with its associate and comes to follow it as a result of WH-movement to the edge of the DP (see also Horrocks & Stavrou 1987; McCloskey 2000; Coon 2009; Ewing 2022; Doliana 2022 and Zyman 2022, fns. 3-4).

(26) *Pira: Quantificational Head within the NP*

[_{QP} WH *pira*_{Q⁰} [_{DP} —]]

When associated with WH-subjects, *pira* can be stranded in the exact set of positions that we should expect to host the absolutive DP. First, it can always surface immediately after the base position of its associate. When it quantifies the ditransitive GOAL, then, it can surface at the right edge of the vsOD string (27a), before adjuncts to the *voiceP*. When it quantifies an absolutive EXT, in turn, it can precede indefinite INTs (27b).

(27) *Pira: Stranding in Thematic Positions*

- a. [_{CP} Innai [_{POLP} na-be-ngang iKaco' bunga — **pira**] digena']]?
 who 3ERG-give-APPL NAME flower all earlier
 ‘Who all did Kacho’ give flowers to earlier?’
- b. [_{CP} Innai [_{POLP} mam-be-ngang — **pira** ande tau] diting]]?
 who ANTIP-give-APPL all food people there
 ‘Who all is giving people food over there?’

This type of stranding then extends in a similar way up the path of the absolutive DP. When the absolutive argument is the GOAL, for instance, it typically raises to SPEC,_{VP} and then to SPEC,*voiceP* (13). When *pira* associates with an absolutive GOAL, it can be stranded in each of these positions: falling just above the INT (28a) or above verbs in *voice*⁰ (28b).

(28) *Pira: Stranding in Intermediate Positions*

- a. Innai byasa [_{voiceP} na-be-ngang [_{VP} iKaco' — **pira** [_{applP} bunga —]]] dini?
 who usually 3ERG-give-APPL NAME all flower here
 ‘Who all does Kacho’ usually give flowers to here?’

- b. Innai byasa [_{voiceP} — **pira** na-be-ngang [_{VP} iKaco' — bunga —]] dini?
 who usually all 3ERG-give-APPL NAME flower here
 ‘Who all does Kacho’ usually give flowers to here?’

It is also impossible for *pira* to take positions that cannot host the absolutive DP. As the absolutive DP raises directly to SPEC,*voiceP* from an inner SPEC,*VP*, for instance, it cannot surface between the verb and the EXT when the verb remains in *voice*⁰ (10a). *Pira*, too, cannot appear between the EXT and an unraised *v* (29a). But when the verb raises, the pattern is reversed (29b)—corresponding to the shift in the position of SPEC,*voiceP* (10b).

(29) *Pira: Only Strands Along the Path of the Associate*

- a. *Innai byasa [_{voiceP} na-alli-ang [_{VP} — **pira** iKaco' — [_{applP} —]]] dio?
 who usually 3ERG-buy-APPL all NAME there
 INTENDED: ‘Who (*all) does Kacho’ usually buy them for there?’
- b. Innai na-alli-ang [_{voiceP} — **pira** [_{VP} iKaco' — [_{applP} —]]] dini?
 who 3ERG-buy-APPL all NAME there
 ‘Who all does Kacho’ usually buy them for here?’

The result is a straightforward argument for WH-QUANTIFIER STRANDING (Sportiche, 1988; Shlonsky, 1991; Merchant, 1996; McCloskey, 2000; Bošković, 2004; Henry, 2012; Zymann, 2018; Doliana, 2022): *pira* forms a constituent with the absolutive WH-phrase, raises with it along its path of movement, and can be realized in low positions along that chain. On this view, we can leverage its distribution to trace the path of extraction: as it can appear in SPEC,*VP* and SPEC,*voiceP*, WH-subjects must pass through these positions too.²

(30) *Stranding Reveals the Path of Subject Extraction*

- a. *Pira* POSITIONS: [_{CP} WH ... [_{voiceP} (*pira*) [_{VP} (*pira*) [_{applP} (*pira*)]]]]
- b. ABSOLUTIVE WH-MVT: [_{CP} WH ... [_{voiceP} _____ [_{VP} _____ [_{applP} _____]]]]

²The facts of prosody suggest a particular analysis of this effect. *Pira* always falls after the right edge of a maximal phonological phrase Itô & Mester 2007, 2012, 2013. It thus seems reasonable to assume that its position is established via scattered deletion (Fanselow & Ćavar, 2003): *pira* moves along the full path of WH-movement and one link along its chain is selected as the optimal position for spell-out in the phonology..

Turning now to the space above the *voiceP*, we can observe that it is impossible for *pira* to be stranded anywhere in the string of string of auxiliaries (31). This fact suggests that WH-subjects never pass through positions between SPEC,*voiceP* and SPEC,ASPP.

(31) *Pira-Stranding: No Landing Sites Between SPEC,voiceP + SPEC,ASPP*

- a. Innai [POLP **ndang** [AUXP1 (*pira) **rua** [AUXP2 (*pira) **yari** [voiceP — lamba]]]]?
 who not all ever all DID go
 INTENDED: ‘Who all never DID go?’
- b. Innai [POLP **byasa** [AUXP1 (*pira) **pura** [voiceP — ma’jama] wattummu malai]]?
 who usually all done work when you return
 INTENDED: ‘Who all is usually done working by the time you come home?’

In the subject WH-questions where temporal adverbs raise up to SPEC,TNSP, however, *pira* can always be stranded in SPEC,ASPP. This pattern is licit no matter the base position of the absolutive DP: like every pattern of intermediate stranding, it can emerge under extraction of the transitive INT (32a), antipassive EXT (32b), and ditransitive GOAL.

(32) *Pira: Stranding in the Intermediate Subject Position*

- a. Innai [TNSP diruambongi — **pira** [POLP byasa [voiceP — na-ita iKaco’]]]?
 who in the past all usually 3ERG-see NAME
 ‘Who all did Kacho’ usually see in the past?’
- b. Innai [TNSP diwongi — **pira** [POLP man-dundu [voiceP — ballo’]]]?
 who last night all ANTIP-drink palm wine
 ‘Who all was drinking palm wine last night?’

This fact shows that these WH-subjects do not extract directly from SPEC,*voiceP*: rather, they must at least pass through SPEC,ASPP. The result is that this Anti-Agreement Effect must emerge from some alternative path to subject extraction: one which requires WH-subjects to interact with all the usual heads of the extended TP (Ouhalla 1993; Schneider-Zioga 1995; after Chomsky 1981; Pesetsky 1982; Rizzi 1982) or one that forces them to raise through SPEC,TNSP but somehow evade interaction with τ^0 (Shlonsky 2014; Pesetsky 2019; see also McCloskey 2000; den Dikken & Griffiths 2018; Messick 2020).

2.4 Interim Summary

These results allow us to recast the Anti-Agreement Effect as a circumscribed breakdown that surrounds the heads c^0 and τ^0 . Rather than reflecting a much wider disruption that plays out across the middle field, such as the suspension of A-movement past *SPEC,voiceP*, I propose that it emerges from the interaction in (33): when the absolutive DP reaches its usual A-position of *SPEC,TNSP*, it canonically interacts with τ^0 , but this type of interaction is suspended in the exact set of contexts where it seems to interact with c^0 .

(33) *Anti-Agreement: ✗Interaction with Both T and C*

- a. REGULAR FEEDING:
$$\left[{}_{\text{TP}} \text{DP}_{\text{ABS}} \quad \tau^0 \quad \left[{}_{\text{TNSP}} \text{DP}_{\text{ABS}} \quad \dots \right] \right]$$
- b. SUBJECT EXTRACTION:
$$\left[{}_{\text{CP}} \text{WH}_{\text{ABS}} \quad \left[{}_{\text{TP}} \text{ — } \tau^0 \quad \left[{}_{\text{TNSP}} \text{WH}_{\text{ABS}} \quad \dots \right] \right] \right]$$

On this analysis, our Anti-Agreement Effect begins to resemble a wider set of patterns that seem to implicate analogous restrictions on simultaneous interaction with τ^0 and c^0 (Chomsky, 2008): for instance, the apparent ban on movement from *SPEC,TP* to *SPEC,CP* (Lasnik & Saito 1992; Erlewine 2016, 2020). The challenge that unfolds before us, then, is to work out an analysis of Anti-Agreement that is grounded in the derivational relationship between the heads of the TP and CP: one that explains why WH-subjects raise to the usual subject position and then fail to interact with τ^0 when they are bound to interact with c^0 . To this end, we can move forward with the following questions in hand.

(34) *Guiding Questions of the Investigation*

1. What is the mechanism that derails the usual behavior of the head τ^0 ?
2. How can this mechanism be triggered by A-movement to *SPEC,TNSP*?
3. Why does this interaction emerge at the specific level of *SPEC,TNSP*?

3 Subject Extraction and Clause Reduction

We thus embark on the second arc of our investigation with a question about the fate of τ^0 . Many analyses of Anti-Agreement begin from a particular hypothesis about structural stability: in contexts of subject extraction, the head that hosts subject agreement continues to project (Ouhalla, 1993; Schneider-Zioga, 1995, 2000, 2007; Cheng, 2006; Diercks, 2009). This assumption, however, is frequently grounded in a particular fact of the morphology: in Trentino, Fiorenino, and many varieties of Bantu and Berber, subject extraction triggers the replacement of regular agreement with a type of morphological default, rather than its total loss. The task of this section, then, is to ask whether the opposing pattern in Mandarin reflects anything more than a morphological quirk: if subject extraction forces the total loss of agreement, might it also implicate the syntactic suppression of τ^0 ?

3.1 The Structure of the Mandarin CP

To tackle this question, we can turn to the wider connection from subject extraction to clause reduction—and more specifically, a series of effects to suggest the derivational elimination of the heads τ^0 and c^0 . We can launch this next arc of our investigation from the syntax of long-distance extraction, which differs from that of short subject extraction in one key respect: it proceeds from clauses that should otherwise host overt elements in c^0 .

Mandarin has a finite complementizer *mua'*, derived from *k<um>ua* “ANTIPASSIVE-say” and cognate to the Selayarese *ko* (Finer, 1997), which surfaces beneath most embedding verbs. The typical pattern of embedding is shown in (35): the matrix verb is transitive, the matrix τ^0 hosts third-person agreement, and *mua'* falls at the top of the embedded clause.

(35) *The Complementizer Mua'*

- a. U-issang i [_{CP} **mua'** [_{TP} pole bo i iCicci']].
 1ERG-know 3ABS that come again 3ABS NAME
 ‘I knew that Chichi’ was coming back.’

- b. Tapi' ndang i na-pissangngi [_{CP} **mua'** [_{TP} pole to i iKaco']].
 but not 3ABS 3ERG-announce that come also 3ABS NAME
 'But she didn't announce that Kacho' was coming too.'

The embedded cps headed by *mua'* show all the usual functional structure of the TP: they strictly host absolutive agreement, they can host auxiliaries and high adverbs in SPEC,TNSP (36a), and they show overt movement of absolutive QPs to SPEC,TNSP (36b).

(36) *The Complementizer Mua': Embeds TPs*

- a. Na-irrangngi i [_{CP} **mua'** [_{TP} [_{TNSP} diruambongi i [_{POLP} rua latto' setang]]]].
 3ERG-heard 3ABS that in the past 3ABS once appear demon
 'She heard that the demon had once appeared in the past.'
- b. Na-pepinnassai i [_{CP} **mua'** [_{TP} [_{TNSP} ing. setang diwongi i latto']]]].
 3ERG-explain 3ABS that all demon last night 3ABS appear
 'She explained that all the demons appeared last night.'

This c^0 is initially important for the fact that its distribution is tightly constrained. When complement clauses surface beneath transitive verbs, *mua'* is typically required (37a). But the language has a second class of embedding verbs that strictly surface in the antipassive voice, and beneath these verbs, it is typically possible to drop this c^0 (37b).

(37) *Starting Asymmetry: Transitive Verbs Require Overt C*

- a. U-irrangngi i [_{?P} *(**mua'**) malai i].
 1ERG-hear 3ABS that go home 3ABS
 'I heard she was coming home.'
- b. **Ma'**-ua a' [_{?P} (**mua'**) malai i].
 ANTIP-say 1ABS that go home 3ABS
 'I said she was coming home.'

This divide is linked to a specific difference in the external syntax of the complement clause. When complement clauses are selected by antipassive verbs, the matrix syntax of subjecthood singles out the antipassive EXT: that argument agrees with τ^0 and raises

along the usual path of the absolutive DP. This fact is then matched by a restriction in the domain of Condition C: beneath antipassive verbs, complement clauses cannot contain R-expressions that corefer with an absolutive pronominal EXT of the embedding verb (38). If these clausal complements remain in the VP, this restriction follows neatly from the fact that the absolutive EXT of the embedding verb should c-command the embedded clause.

(38) *Antipassive Clauses: Absolutive EXT > Embedded CP*

- a. **Ma'**-ua i *pro*_{EXT} [_{?P} ____ malai i kindo'-na iCicci' manini].
 ANTIP-say 3ABS she (that) go home 3ABS mom-3GEN NAME later
 'She_i said that Chichi's_{*i,j} mom will go home later.'
- b. **Mang-**akui i *pro*_{EXT} [_{CP} mua' na-sayang o kotta'-na iCicci' diolo'].
 ANTIP-admit 3ABS she that 3ERG-love 2ABS boyfriend-3GEN NAME first
 'She_i admitted that Chichi's_{*i,j} boyfriend loved you first.'

In transitive contexts where absolutive agreement seems to target the embedded CP, this restriction disappears. The following examples show this effect: when clausal complements are introduced by transitive verbs like *irrangngi* "hear" and *pissangngi* "announce," it is possible for a pronominal EXT of the embedding verb to corefer with any R-expression in the embedded clause: for instance, in the absolutive DP (39a) or the ergative DP (39b).

(39) *Transitive Clauses: Ergative EXT < Embedded CP*

- a. **Na-**irrangngi i *pro*_{EXT} [_{CP} mua' malai i kindo'-na iCicci' manini].
 3ERG-hear 3ABS she that go home 3ABS mom-3GEN NAME later
 'She_i heard that Chichi's_{i,j} mom will go home later.'
- b. **Na-**pissangngi i *pro*_{EXT} [_{CP} mua' na-ita o kindo'-na iCicci' digena'].
 3ERG-announce 3ABS she that 3ERG-see 2ABS mom-3GEN NAME earlier
 'She_i announced that Chichi's_{i,j} mom saw you earlier.'

This contrast reveals that the A-syntax must raise these complement CPs in the contexts where they seem to trigger agreement with T⁰—raising them above the EXT to SPEC, *voiceP* and likely farther to SPEC, *ASPP* and SPEC, *TNSP*. Setting aside the matter of whether these steps are contingent upon encasement in a covert DP (Rosenbaum 1965; McCloskey 1991;

Davies & Dubinsky 2001; cf. Bošković 1995), the result is the syntax in (40): beneath transitive verbs but not their antipassive counterparts, complement CPs are sentential subjects.

(40) *The External Syntax of Complement Clauses*

- a. TRANSITIVE: $\left[\text{TP} \text{ CP}_{\text{ABS}} \left[\text{POLP} \text{ V}_{\text{TRANS}} \left[\text{voiceP} \text{ DP}_{\text{ERG}} \text{ —CP} \right] \right] \right]$
- b. ANTIPASSIVE: $\left[\text{TP} \text{ DP}_{\text{ABS}} \left[\text{POLP} \text{ V}_{\text{ANTIP}} \left[\text{voiceP} \text{ —DP} \text{ CLAUSE}_{\text{OBL}} \right] \right] \right]$

We thus arrive at a link from the possibility for movement to the visible presence of c^0 : when complement clauses raise along the path of the absolutive DP and interact with τ^0 , they must overtly host *mua'*. To capture this asymmetry, I propose that the systems of agreement and A-movement (or, alternatively, an encasing D^0) are sensitive to the presence of c^0 . On this view, we can understand the surface distribution of *mua'* to follow from the fact that it is the only possible exponent of the finite c^0 : in other words, sentential subjects are forced to host *mua'* because the Mandar lexicon lacks a null counterpart of this c^0 . The result is a systematic structural divide between the clauses that host *mua'* and those that do not (and see Doherty 1993, 2000 on English): the clauses that host *mua'* are CPs (41a) and the clauses that lack it are smaller (41b). For concreteness, I take them to be TPs.

(41) *Complement Clauses: Differences in Structural Size*

- a. OVERT C: $\left[\text{CP} \text{ mua}' \text{ TP} \right]$
- b. BARE TP: $\left[\text{TP} \text{ —} \right]$

3.2 The Syntactic Suppression of C

From this position, we can now establish an interaction between subject extraction and c^0 . The clauses that host *mua'* are transparent to non-subject extraction. When adjunct WH-phrases are generated within them, then, they undergo long-distance WH-movement in a regular way: they not trigger Anti-Agreement and they freely cross the overt c^0 (42).

(42) *Mua’-Clauses: Transparent to Adjunct Extraction*

Pirang i mu-irrangngi [_{CP} **mua’** latto’ i do setang o ____]?
 when 3ABS 2ERG-hear that appear 3ABS that demon there

‘When did you hear that that demon there appeared?’

Embedded subject extraction, however, forces the surface suppression of c^0 . *Mua’* must disappear in all the embedded clauses that launch extraction of the absolutive DP (43a), even though it is typically required in the clausal complements of transitive verbs. This pattern is then matched by the corresponding disappearance of the embedded τ^0 (43b): in other words, our Anti-Agreement Effect is matched by a Comp-Trace Effect.³

(43) *Long-Distance Subject Extraction: Suppression of C*

a. Innai mu-irrangngi [_{?P} ____ u-ita ____ dionging]?
 who 2ERG-hear (that) 1ERG-see yesterday

‘Who did you hear that I saw yesterday?’

b. Innai mu-sa’ding [_{?P} ____ na-pile ____/(*) ____ marondong]?
 who 2ERG-feel (that) 1ERG-see 3ABS yesterday

‘Who do you feel that they’ll pick tomorrow?’

The facts of quantifier stranding then reveal that the Comp-Trace Effect is linked to extraction from the usual subject position. In contexts of long-distance extraction, *pira* can be stranded in every position on the path of the absolutive DP up to SPEC,TNSP (44b).

³On the surface, the language allows WH-dependencies that appear to stretch into the subject positions of embedded clauses that visibly host this c^0 (1). In this context, however, the embedded clause never shows Anti-Agreement and *pira* cannot be stranded along the expected path of extraction in the *mua’*-CP. I take these facts to suggest that *mua’*-clauses are opaque to subject extraction (and see [Finer 1997](#) for a parallel claim about *ko*-CPs in Selayarese): in this context, WH-phrases originate in the embedded SPEC,CP and bind resumptive pronouns lower in the CP ([Borer 1981](#); [McCloskey 1990](#); [Asudeh 2004](#); [Sichel 2014](#); [Hewett 2023](#)).

(1) *Mua’-Clauses: Opaque to Subject Extraction*

Innai na-irrangngi [_{CP} **mua’** lamba i ____ (**pira*)]?
 who 3ERG-hear that go 3ABS all

INTENDED: ‘Who all did she see that ____ did go?’

(44) *Long-Distance Subject Extraction: Evidence for Direct Movement*

- a. Innai mu-sa'ding [_{TNSP} marondong [_{ASPP} — **pira** na-pile]]?
 who 2ERG-feel tomorrow all 3ERG-pick
 'Who all do you feel that he'll choose tomorrow?'
- b. Innai mu-irrangngi dionging [_{TNSP} — **pira** marondong na-pile]?
 who 2ERG-hear yesterday all tomorrow 3ERG-choose
 'Who all did you hear yesterday that he'll choose tomorrow?'

These effects are then matched by two further shifts to reveal that subject extraction derails the construction of the usual projection CP. When embedded clauses launch subject extraction, to begin, they cease to trigger third-person agreement on the matrix T⁰ (45a): a fact which suggests that they are smaller than the clauses that host overt exponents of c⁰. This shift is then mirrored by a parallel realignment in the system of Condition C: beneath transitive verbs, the embedded clauses that launch subject extraction lose the ability to host R-expressions coindexed with a pronominal matrix EXT (45b): a fact which suggests that they cannot become sentential subjects and raise along the path of the absolutive DP.

(45) *Embedded Subject Extraction: Derails Construction of the Embedded CP*

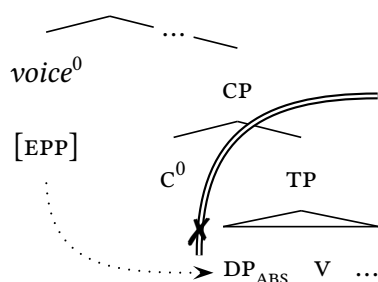
- a. Innai mu-irrangngi ___/*(i) diruambongi [_{?P} — na-sayang iKaco']?
 who 2ERG-hear 3ABS in the past 3ERG-love NAME
 'Who did you hear in the past that Kacho' loved?'
- b. Innai na-irrangngi *pro*_{EXT} [_{CP} na-pelambi'i kindo'-na iCicci' digena']?
 who 3ERG-hear she 3ERG-visit mom-3GEN NAME earlier
 'Who did she_i hear that Chichi's_{*i,j} mom visited earlier?'

These facts reveal that there is a syntactic shift beneath the surface suppression of c⁰: this particular comp-trace effect reflects a syntactic loss of c⁰ (Bošković 1994, 1996, 1997; Grimshaw 1997), rather than the introduction of a second complementizer that happens to be null and exceptionally licenses subject extraction on a par with the French *qui* (Chomsky & Lasnik 1977; Kayne 1980; Pesetsky 1982; Bennis & Haegeman 1984; Rizzi 1990).

3.3 Interlude: Transparency Effects and τ^0

From this position, we can begin to make a case for the corresponding suppression of τ^0 . This is one that turns on a particular type of transparency effect, and to illustrate, it will be useful to begin from a secondary consequence of the absence of c^0 . When transitive verbs embed CPs, we have seen, the heads of the matrix VP cannot reach into the embedded TP: the matrix *voice*⁰, for instance, must attract the CP and not the embedded absolutive DP. The result is a type of opacity that we can attribute to PHASE THEORY (Chomsky 2001; though cf. Halpert 2019): the head c^0 defines a cyclic domain of syntactic computation, and as a result, the material in its complement becomes inaccessible to all higher heads.

(46) Phase-Based Locality



In the clauses that lack c^0 , this restriction disappears: when *mua*' is suppressed, the embedded absolutive DP becomes accessible to the matrix v^0 . We can see this fact with a turn to a second class of transitive verbs that select complements without c^0 : *sanga* “think” and *hara*’ “hope.” These verbs embed complements that project up to the TP, hosting absolutive agreement and the positions we have identified as SPEC,ASPP and SPEC,TNSP.

(47) A Second Class of Embedding Verb

- a. **U-sanga** i [_{TP} _____ **lambda i** **iCicci'** marondong].
 1ERG-think 3ABS (*that) go 3ABS NAME tomorrow
 ‘I think (*that) Chichi’ will go tomorrow.’
- b. **Mu-sanga** i [_{TP} [_{TNSP} **diongning i** [_{ASPP} **iCicci'** [_{POLP} **di-ita** **diò**]]]]?
 2ERG-think 3ABS yesterday 3ABS NAME PASS-see there
 ‘Do you really think that yesterday Chichi’ was seen THERE?’

In keeping with the fact that they lack c^0 , these clauses are transparent to the matrix T^0 : in the matrix clauses built around *sanga* and *hara'*, T^0 agrees with the embedded absolutive DP (48a). It cannot show the rigid third-person agreement that emerges around sentential subjects, which ban agreement between the matrix T^0 and the lower absolutive DP (48b).

(48) *Complement Clauses without Mua': Transparent to Agreement*

- a. U-sanga o/*i [_{TP} lamba o i'o marondong].
 1ERG-think 2ABS/3ABS go 2ABS you tomorrow
 'I thought you were going tomorrow.'
- b. U-irrangngi i/*o [_{CP} mua' lamba o i'o marondong].
 1ERG-hear 3ABS/2ABS that go 2ABS you tomorrow
 'I heard that you were going tomorrow.'

This fact then correlates with explicit evidence for higher raising of the absolutive DP. Unlike sentential subjects, the clausal complement of *sanga* and *hara'* remain in the VP, and we can see this fact in the system of Condition C: in the embedded clause, R-expressions within constituents like the ergative DP cannot freely corefer with a pronominal matrix EXT (49a). But this restriction is lifted around the embedded absolutive DP (49b).

(49) *Complement Clauses without Mua': Binding Facts Suggest Raising*

- a. Na-sanga i pro_{ERG} [_{TP} na-ita i kindo'-na iCicci' digena'].
 3ERG-think 3ABS she 3ERG-see 3ABS mom-3GEN NAME earlier
 'She_i thinks that Chichi's_{i,j} mom saw it earlier.'
- b. Na-sanga i pro_{ERG} [_{TP} lamba i kindo'-na iCicci' digena'].
 3ERG-think 3ABS she go 3ABS mom-3GEN NAME earlier
 'She_i thinks that Chichi's_{i,j} mom left earlier.'

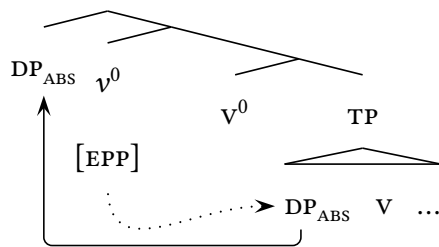
When we introduce quantifiers like *inggannanna* into these embedded absolutive DPs, moreover, we see the explicit raising in (50a): the embedded absolutive DP is forced to move to the matrix clause and cannot remain in the embedded clause (50b).

(50) *Complement Clauses without Mua': Visible Raising of Absolutive QPs*

- a. **Ing. sola-u** u-sanga i [_{?p} mu-ita i ___ marondong].
 all friend-1GEN 1ERG-think 3ABS 2ERG-see 3ABS tomorrow
 'I think you'll see all my friends tomorrow.'
- b. *U-sanga i [_{?p} **ing. sola-u** mu-ita i ___ marondong].
 1ERG-think 3ABS all friend-1GEN 2ERG-see 3ABS tomorrow
 INTENDED: 'I think you'll see all my friends tomorrow.'

In the terminology of Ura 1994, the result is a type of hyperraising (for review: Zyman 2023; also Alexiadou & Anagnostopoulou 1999; Uchibori 2001; Ferreira 2000, 2004, 2009; Martins & Nunes 2005, 2009, 2010; Zeller 2006; Deal 2017; Halpert 2019; Lee & Yip 2024): *sanga* and *hara'* select fully finite TPs and then force raising of the embedded absolutive DP. Setting aside the matter of ACTIVITY (Chomsky, 2001), we can understand this pattern along much the same lines as Zyman 2017 and Fong 2019 (and see Tanaka 2002; Şener 2008; Takeuchi 2010; Alboiu & Hill 2016; Bondarenko 2017; Wurmbrand 2019): the matrix v^0 can attract the embedded absolutive DP because it falls at the edge of the embedded TP (suggesting that the absolutive DP must then consistently raise past SPEC,TNSP to SPEC,TP).

(51) *Hyperraising: Absolutives Accessible at the Edge of the TP*



Shifting one level down, we can extend the same analysis to a class of complements that seem to project no farther than the TNSP. Mandarin has a class of regular raising verbs that includes *mala* “be able,” *minnassa* “be clear,” several aspectual predicates (*mamanya* “be in the process of,” *myosa* “stop”), and many transitive verbs (*jolo* “show,” *pokanynyang*

“suspect”). These raising verbs embed complements that lack absolutive agreement (52a) but allow adverbs to raise to SPEC,TNSP (52b). I thus propose that they project up to TNSP.

(52) *Raising Complements*

- a. Dionging i **mala** [ASPP pole ___/*i iCicci’ marondong].
 yesterday 3ABS possible come 3ABS NAME tomorrow
 ‘Yesterday it was possible that Chichi’ would come tomorrow.’
- b. Dionging i **minnassa** [ASPP **marondong** pole ___/*i iCicci’ ___].
 yesterday 3ABS clear tomorrow come 3ABS NAME
 ‘Yesterday it was clear that tomorrow Chichi’ would come.’

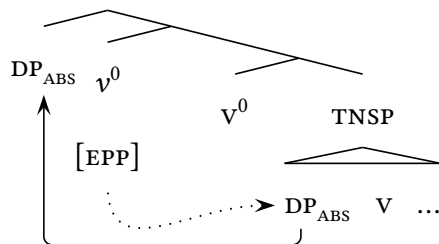
Like the hyperraising complements above, these raising complements are transparent to absolutive agreement: in the clauses built around *minnassa*, the matrix τ^0 must agree with the embedded absolutive DP (53a). They also show all the usual evidence for raising: if the embedded absolutive DP is quantified, it must surface in the matrix SPEC,TNSP (53b).

(53) *Raising Complements: Evidence for Raising*

- a. Dionging o/*i **minnassa** [ASPP pole *pro* marondong].
 yesterday 2ABS/3ABS clear come you tomorrow
 ‘Yesterday it was clear that you would come tomorrow.’
- b. I’o mie’ inggannanna dionging o minnassa [ASPP pole ___].
 you guys all yesterday 2ABS clear come
 ‘Yesterday it was clear that all you guys would come.’

These facts suggest the parallel analysis in (54): embedded absolutive DPs can raise out of these complements because they are always drawn to the edge by attraction to SPEC,TNSP.

(54) *Raising: Absolutes Accessible at the Edge of the TnsP*



3.4 Subject Extraction Triggers the Loss of τ^0

We can now leverage this initial contrast between raising and hyperraising complements to identify a transparency effect linked to τ^0 . This is one that emerges most clearly in the behavior of the adverb *bo* “again,” an element that typically raises to the edge of the TNSP. In matrix clauses, this element follows the first element in the TNSP, on a par with τ^0 (55a). It never takes the usual position of temporal adjuncts at the edge of the *voiceP* (55b).

(55) *The Adverb Bo: Second-Position within the TnsP*

- a. [_{TNSP} Dionging **bo** a' [_{POLP} na-telpong iKaco' __]].
 yesterday again 1ABS 3ERG-call NAME
 ‘Yesterday Kacho’ called me again.’
- b. [_{TNSP} *Dionging __ a' [_{POLP} na-telpong iKaco' **bo**]].
 yesterday 1ABS 3ERG-call NAME again
 INTENDED: ‘Yesterday Kacho’ called me again.’

We can see into the derivational behavior of this adverb with a turn to a second non-finite environment. Mandar has a class of reduced clauses that are selected by the c^0 *mau* “though.” These clauses are able to host auxiliaries, but they lack absolutive agreement—suggesting that they typically only project to POLP (57).

(56) *A Second Class of Non-Finite Clauses: Though-Complements*

- Mau** [_{POLP} ndang rua [_{voiceP} di-ita dini]], masskripsi i de'.
 though not ever PASS-see here dissertate 3ABS allegedly
 ‘Though never seen [–FIN] around here, he’s allegedly dissertating.’

This second environment is useful for the way that it reveals the particular role of ASP^0 . Unlike the complements of control verbs, the non-finite clauses introduced by *mau* are able to host overt absolutive DPs (57a)—suggesting that they provide an exceptional source of structural Case to that DP). But they typically do not allow the absolutive DP to raise past

SPEC,voiceP: when the absolutive DP is quantified, for instance, it must surface beneath all auxiliaries (57b) and cannot raise to the position we have identified as SPEC,ASPP.

(57) *Though-Complements: No Higher Subject Positions*

- a. **Mau** [POLP na-ita [voiceP iCicci' [VP iKaco' ___]] diò],
 though 3ERG-see NAME NAME there
 'Though Kacho' saw [—FIN] Chichi' THERE,'
- b. Mau [POLP pura [voiceP **ing. sola-u** u-ellongngi]],
 though done all friend-1GEN 1ERG-call
 'Though I'm done calling [—FIN] all my friends,'

The situation changes, however, when we force these clauses to project up to the ASPP. Mandarin has a pair of enclitics that introduce distinctions of outer aspect: *mo* “already” and *pa* “yet” (58a). These elements typically follow the highest element in the TNSP, on a par with the enclitics that mark absolutive agreement, and I take them to head the ASPP. When we introduce these elements into the non-finite clauses headed by *mau*, we begin to see a further step of raising for the absolutive DP: when it is quantified, for instance, it starts to raise past the auxiliaries to the position we have identified as SPEC,ASPP (58b).

(58) *Though Complements: The Role of ASP⁰*

- a. Mau [ASPP [POLP pura **mo** [voiceP ___ u-ellongngi]]],
 though done ALREADY 1ERG-call
 'Though I'm already done calling [—FIN] them,'
- b. Mau [ASPP **ing. sola-u** [POLP pura **mo** [voiceP ___ u-ellongngi]]],
 though all friend-1GEN done ALREADY 1ERG-call
 'Though I'm already done calling [—FIN] all my friends,'

The introduction of ASP⁰ triggers an analogous step of movement for the adverb *bo*. In the non-finite clauses that project to POLP⁰, *bo* follows the verb (59a). In the presence of *mo* and *pa*, however, it always raises to follow the highest overt auxiliary (59b).

(59) *The Adverb Bo: Movement to the edge of the AspP*

- a. Mau [POLP yari [voiceP latto **bo** do setang o _]],
 though DID appear AGAIN that demon there
 ‘Though that demon there did appear [–FIN] again,’
- b. Mau [ASPP [POLP yari **bo mo** [voiceP latto _ setan-na _]]],
 though DID again ALREADY appear demon-3GEN
 ‘Though the demon already did appear [–FIN] again,’

When we front temporal adverbs to force the construction of the TNSP, in turn, we see two further steps: absolutive QPs raise to SPEC,TNSP and *bo* raises to follow the adverbs.

(60) *The Adverb Bo: Movement to the edge of the TnsP*

- Mau [TNSP **ing. tau** diwongi **bo** [POLP yari _ [voiceP latto _]]],
 though everyone last night again DID appear
 ‘Though everyone did appear [–FIN] again last night,’

I thus propose that this adverb raises along the path in (61): after entering the derivation as a typical (right-)adjunct to the *voiceP*, it is attracted to an outer SPEC,*voiceP*, then to an outer SPEC,ASPP, and finally to an intermediate SPEC,TNSP. Like absolutive agreement, I assume that it is positioned after the highest element beneath it in the postsyntax.

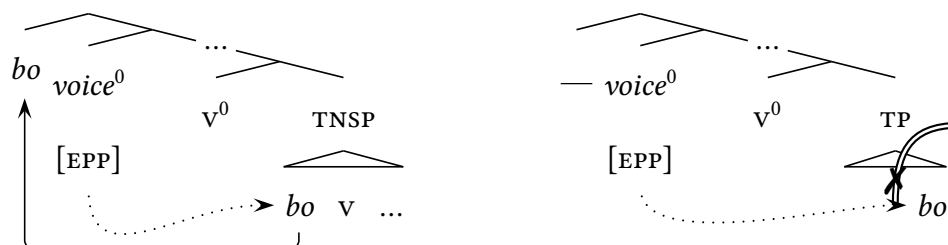
- (61) *The Syntax of Bo “Again”*: [TNSP *bo* [ASP _ [voiceP _ ... _]]]
-

From this position, we can leverage the behavior of this adverb to test for the presence of τ^0 . In a clause that projected up to the level of the TNSP, we might expect to see the pattern of attraction in (62a): if *bo* raises to the edge of the SPEC,TNSP, it should be accessible to the matrix *voice*⁰ and should thus be able to climb into the matrix clause. In clauses that contain structure above the TNSP, however, we might expect to see the pattern reverse: if *bo* does not raise to the edge of the TP, for instance, this type of climbing should be blocked in the clauses that host the head τ^0 (62b), on a dynamic approach to phasehood which takes the highest head along the spine to define a phase (Takahashi, 2010; Bobaljik & Wurmbrand, 2005, 2013; Wurmbrand, 2017; Bošković, 2013, 2014, 2016; Harwood, 2015).

(62) *Transparency for Again-Movement: Predictions*

a. *TnsP = Transparent*

b. *TP = Opaque*



Within the regular system of clausal complementation, these predictions are borne out. When *bo* originates in the TNSP-sized complements of raising verbs, it is unable to remain low: rather, it must raise from the embedded SPEC,TNSP into the matrix clause.

(63) *Raising Complements: Transparent for Bo*

- a. Minnassa **bo** i [ASPP yari — na-tumae].
 clear again 3ABS DID 3ERG-propose
 ‘It’s clear that he did propose to her again.’

- b. *Minnassa i [ASPP yari **bo** na-tumae].
 clear 3ABS DID again 3ERG-propose
 INTENDED: ‘It’s clear that he already did propose to her again.’

When *bo* originates in a hyperraising complement that forms a TP, in turn, the facts are reversed: this adverb must stay in the embedded TP and cannot raise out.

(64) *Hyperraising Complements: Opaque for Bo*

- a. U-sanga i [TP [POLP yari **bo** mi [voiceP na-tumae]]].
 1ERG-think 3ABS DID again ALREADY.3ABS 3ERG-propose
 ‘I think he already did propose to her again.’

- b. *U-sanga **bo** i [TP [POLP yari — mi [voiceP na-tumae]]].
 1ERG-think again 3ABS DID ALREADY.3ABS 3ERG-propose
 INTENDED: ‘I think he already did propose to her again.’

With this much in place, we can now turn to the way that this adverb interacts with the system of WH-movement. In the default case, its placement is not disturbed by the

extraction of non-subjects: when *bo* originates in a TP or CP that launches adjunct WH-movement, for instance, it is strictly trapped in its clause of origin and cannot raise out.

(65) *Non-Subject Extraction: No Effect on Bo*

- a. Pirang i mu-sanga [TP maindong **bo** i [voiceP iKaco'] ___]?
 when 3ABS 2ERG-think escape 3ABS NAME
 'When do you think Kacho' escaped again?'
 b. Myapa i na-pissangngi [CP mua' maindong bo i]?
 how 3ABS 3ERG-announce that escape again 3ABS
 'How did he announce that he would escape again?'

When *bo* originates in a clause that launches subject extraction, however, it is forced to escape: it must raise out and follow the highest element in the matrix TNSP.

(66) *Subject Extraction: Jailbreak*

- a. Innai [TNSP digena' **bo** mu-irrangngi [TNSP di-be-ngang ___ tilang]]?
 who just now again 2ERG-hear PASS-give-APPL ticket
 'Who did you just hear was given a speeding ticket again?'
 b. *Innai [TNSP digena' mu-irrangngi [TNSP di-be-ngang **bo** tilang]]?
 who just now 2ERG-hear PASS-give-APPL again ticket
 INTENDED: 'Who did you just hear was given a speeding ticket again?'

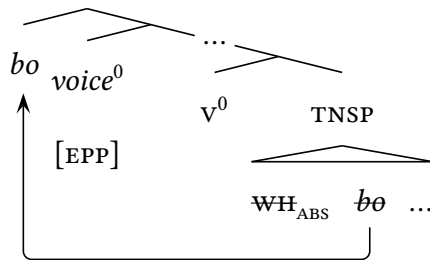
This effect is then mirrored in the behavior of many further elements that typically cluster after the first element in the TNSP. Unstressed pronouns and demonstratives, for instance, move along the same path as *bo* and invariably remain within the TNSP when they originate in embedded TPs (67a). But when they originate in clauses that launch subject extraction, they strictly raise to follow the first element in the matrix TNSP (67b).

(67) *Subject Extraction: More of the Jailbreak*

- a. Apana mu-sanga i [TP dionging i **yau do** u-be-ngang iCicci' o]?
 Why 2ERG-think 3ABS yesterday 3ABS 1SG that 1ERG-give-APPL NAME PRT
 'Why do you think that yesterday I gave that to Chichi?'
 b. Innai [TNSP digena' **yau do** mu-irrangngi [TNSP u-be-ngang ___ ___ o]]?
 who just now 1SG that 2ERG-hear 1ERG-give-APPL PRT
 'Who did you just hear that I gave that to?'

These facts establish a tight parallel between the clauses that launch subject extraction and those that launch raising: both are transparent for the attraction of *bo* and other second-position elements. This result, in turn, suggests the pattern of reduction in (68): the clauses that launch subject extraction must lack τ^0 and only project to the TNSP.

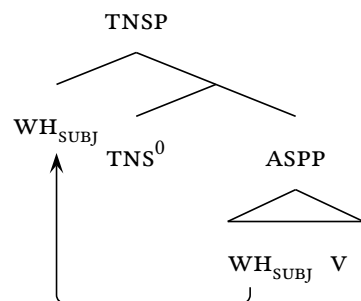
(68) *Subject Extraction: Bundling Past the TP*



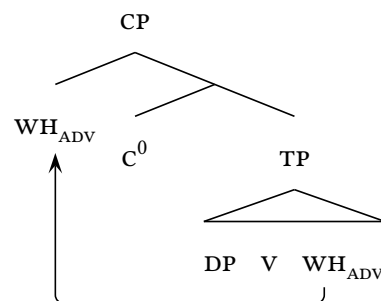
This analytical step, in turn, allows us to draw out and sharpen the theoretical targets that our analysis of Anti-Agreement will have to meet. To begin, it establishes that subject extraction must force the syntactic suppression of the head that canonically hosts subject agreement—tracing both the morphological shape of this effect and the theoretical thrust of Shlonsky 2014 and Pesetsky 2019. Second, it suggests that this pattern of suppression must reach down to an A-position that hosts the subject in the extended TP: the clauses that launch subject extraction must project exactly to the TNSP. In tandem with this result, at last, it suggests that the mechanism beneath subject extraction must be distinct from the regular $\bar{A}$ -movement that carries non-subject WH-phrases to SPEC,CP (Chung & McCloskey 1983; Chomsky 1986; see also Bošković 2024): rather, it would seem to be a type of A-movement that suspends structure-building above SPEC,TNSP.

(69) *Subject Extraction = A-Movement to SPEC,TNSP*

a. *Subject Extraction*



b. *Non-Subject Extraction*



4 The Unfolding Framework

To capture this network of interactions in derivational terms, then, I propose that the heads that are suppressed in contexts of subject extraction in Mandar must be constructed within the derivation by a process that is derailed by A-movement of WH-subjects to SPEC,TNSP. Drawing from Rizzi 1997, more specifically, I propose that the Mandar τ^0 and c^0 are canonically introduced to the derivation within a bundled head (Giorgi & Pianesi, 1997): one that hosts the featural content of c^0 , τ^0 , and TNS^0 . Following Shimada 2007, I will argue, this head type of head is typically forced to UNFOLD through a cyclic process of excorporation and reprojection, yielding the independent sequence of heads TNS^0 - τ^0 - c^0 . In the presence of WH-subjects, however, I will argue that this process is suspended by a derivational pressure for multitasking (Pesetsky & Torrego, 2001; Van Urk & Richards, 2015; Newman, 2024): one that leverages the WH-phrases drawn up by A-features to SPEC,TNSP in order to parasitically check the $\bar{A}$ -features that typically surface at the level of the CP. The result is an analysis that derives the suspension effect from the regular interaction of two component parts: a universal system of UNFOLDING and a language-particular pattern of head-bundling that cuts across the extended TP and CP.

4.1 Head-Bundling

The analysis begins from the framework of bundling developed by Giorgi & Pianesi 1997. To account for cross-linguistic variation in the shape of clausal spine, Giorgi & Pianesi 1997 propose a featural reformulation of the types of functional sequence put forward by Grimshaw 1991 and Cinque 1999. The central ingredients of this system are presented in (70): the heads along the clausal spine project from a sequence of features that are organized into a cross-linguistically universal order (70a), but they are packaged into bundled

heads in a language-particular way, such that each feature has the capacity to head its own projection or bundle together, in the lexicon, with any number of additional features that form a contiguous span along the universal hierarchy (70b).

(70) *Background: The Feature-Scattering Framework*

a. THE UNIVERSAL ORDERING CONSTRAINT

Features are ordered so that given $F_1 > F_2$, the checking of F_1 precedes the checking of F_2 . Giorgi & Pianesi 1997, 14

b. THE FEATURE-SCATTERING PRINCIPLE:

Each feature can head a projection [or, alternatively, bundled together with features adjacent along the universal hierarchy]. Giorgi & Pianesi 1997, 15

This framework provides a natural way to capture many types of cross-linguistic variation, and it has since been widely employed to capture such patterns in the extended CP (Manetta, 2011; Bennett *et al.*, 2012; Hsu, 2017, 2021). Rizzi 1997, for instance, proposes that this type of bundling lies behind the Comp-Trace Effect in English: in that language, he argues, subject extraction requires the emergence of a bundled c^0 that hosts the features that typically define $FORCE^0$ and FIN^0 (71a). This pattern of bundling has two effects: it prevents the insertion of the canonical exponent of FIN^0 , thus forcing the presence of the null c^0 , but it renders the content of $FORCE^0$ accessible to external selection, thus allowing these clauses to be embedded beneath verbs that select full CPs. Extending the same analysis to Mandarin, then, we might posit the analogous pattern in (71b): subject extraction requires a bundled head that contains the features that define TNS^0 , T^0 , and c^0 .

(71) *Subject Extraction: Clause Reduction*

- a. *English*: $[_{FORCE/FINP} WH_{SUBJ} FORCE/FIN^0 [_{TP} \text{ — } \dots v]]$
- b. *Mandar*: $[_{C/T/TNSP} WH_{SUBJ} C/T/TNS^0 [_{ASPP} \text{ — } \dots v]]$

4.2 Derivational Structure-Building

To understand how this type of bundling might be derivationally forced by the A-movement of WH-subjects, we can begin with the mirror image of our effect: cases where the syntax seems to unfold complex heads in order to trigger WH-movement of non-subjects. The classical pattern of this type involves head-movement from τ^0 to c^0 (Emonds, 1976; Den Besten, 1983; Koopman, 1984), which is most familiar from the systems of non-subject WH-movement in English (Koopman, 1983; Pesetsky & Torrego, 2001) and French (Kayne 1983; Rizzi & Roberts 1989). To understand the proposed role of this step, we can begin from the way that it manifests in German and Dutch—where a major strand of work argues that non-subject attraction forces further structure-building at the level of the CP. Travis 1984 and Zwart 1997 propose that the clauses which satisfy the verb-second requirement by attracting subjects project up to the level of the TPs, while those that satisfy the v2 requirement by drawing up non-subjects project up to the level of the CPs (see also Hoekstra & Marác 1989; Branigan 1996; Fanselow 2002; Bošković 2024). To capture this asymmetry, I propose that the derivation forces the construction of the CP layer in order to attract non-subjects to the v2 position: put differently, non-subject attraction forces unfolding of a complex c/τ^0 (see also Bennett *et al.* 2012; Martinović 2015; Erlewine 2016).

(72) *Verb-Second: Structural Asymmetry*

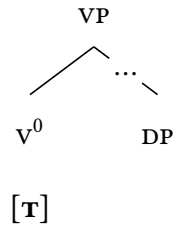
- a. SUBJECT v2 CLAUSES: $[_{TP} \text{ DP}_{SUBJ} \quad V \quad \dots]$
- b. NON-SUBJECT v2 CLAUSES: $[_{CP} \text{ XP}_{NONSUBJ} \quad V \quad \dots [_{TP} \text{ DP}_{SUBJ} \quad \dots]]$

This type of structure-building head-movement is typically held to be triggered by category features that are lexically specified to raise to the root and REPROJECT, in the tradition of Holmberg 1991 and Ackema *et al.* 1993 (see also Koenenman 2000; Bury 2003; Fanselow 2004; Surányi 2005; Shimada 2007; Georgi & Müller 2010; Martinović 2015; Erlewine 2018). Within a different derivational context, for instance, Ackema *et al.* 1993

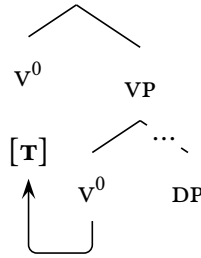
and Fanselow 2004, 2009, for instance, propose that the same type of head-movement is responsible for the construction of the TP. The essentials of their account are sketched in (73): when verbs are merged, they discharge their selectional features, project the VP (73a), and then turn to a REPROJECTION FEATURE denoted here as $[\underline{\mathbf{T}}]$. This feature forces syntactic head-movement to the root (73b), in a step that resembles head-movement to a specifier position (Toyoshima, 2001; Matushansky, 2006; Harizanov, 2019). But it passes up a label, making it the head of the new constituent built at the root. When this type of head-movement is triggered by $[\underline{\mathbf{T}}]$, then, it derivationally constructs a TP (73c).

(73) *Head-Movement and Reprojection*

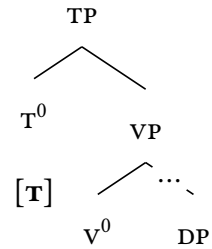
a. *Starting Point*



b. *Head-Movement*

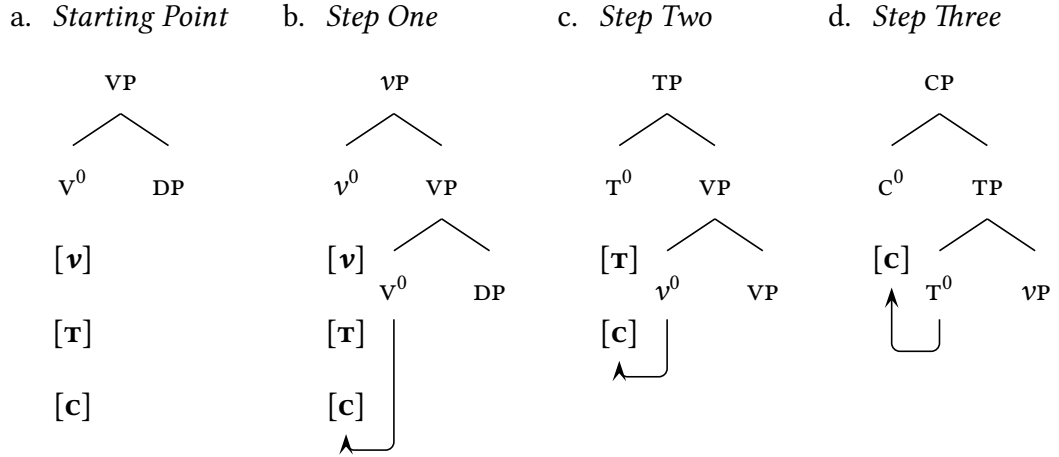


c. *Reprojection*



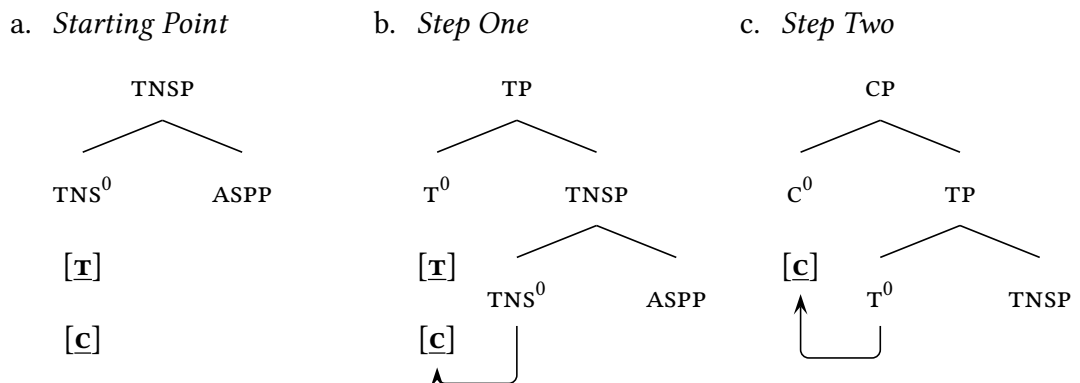
Ackema *et al.* 1993 and Fanselow 2004, 2009 propose that this type of head-movement raises the entire content of v^0 to T^0 , ruling out any morphological separation between the heads that carry reprojection features and the reprojected heads that they create. This stance fits neatly with the lexicalist position that they adopt, but it stands at odds with the hypothesis that patterns of exponence can be rerouted by structure-building operations of this type (a stance implicit in both Rizzi 1997 and Giorgi & Pianesi 1997). To resolve this tension, then, I follow Shimada 2007 in taking this type of head-movement to implicate a cyclic process of excorporation and reprojection: one that raises successively smaller pieces of this bundled to the root to construct new heads that can be independently exponed. The core of this framework is sketched below: pushing the feature-scattering theory to its maximum extent, Shimada 2007, proposes that the entire clausal spine is gradually unfolded from a single bundle that hosts the features $[\mathbf{v}]$, $[\mathbf{v}]$, $[\mathbf{T}]$, and $[\mathbf{c}]$.

(74) *Shimada: Bundling and Excorporation*



To derive the relationship between TNS^0 , T^0 , and c^0 , we can imagine that the same process of unfolding targets the bundled $\text{TNS}/\text{T}/\text{c}^0$. I assume that the Mandar lexicon contains independent heads that correspond to TNS^0 and T^0 , selected in the numerations that construct reduced complements that only project to the TNSP and TP . To build the finite clauses that project to the CP , however, I propose that the numeration invariably selects the bundle $\text{TNS}/\text{T}/\text{c}^0$. This is a head that hosts the reprojection features in (75a): $[\underline{\text{T}}]$, which forces unfolding to build the TP , and $[\underline{\text{c}}]$, which forces unfolding to build the CP . As these features are ordered extrinsically along the universal functional sequence of [Giorgi & Pianesi 1997](#), they must be derivationally ordered in this head (in the sense of [Heck & Müller 2007](#)): once $\text{TNS}/\text{T}/\text{c}^0$ is merged as the sister to ASP^0 , $[\underline{\text{T}}]$ must force unfolding to build the TP (75b) and $[\underline{\text{c}}]$ must then trigger the same process to build the CP (75c).

(75) *Mandar: Unfolding from Asp to C*



To lay the groundwork for interaction between unfolding and attraction, I propose that these category features are interspersed with the derivational features that drive *MOVE* and *AGREE*. In bundled heads like $\text{TNS}/\text{T}/\text{C}^0$, these features must be derivationally ordered in a language-particular and lexically determined way: the feature that attracts *bo* to an intermediate SPEC,TNSP must fall just after the feature that attracts temporal adverbs, while the feature that draws absolutive *DPS* to SPEC,TNSP must fall just before $[\underline{\text{T}}]$. Denoting movement-driving features with bullets ($[\bullet\text{x}\bullet]$) and ϕ -agreement features with asterisks ($[*\phi*]$), after Heck & Müller 2007, we can thus summarize the content of this head as (76).

(76) *The Integrated Feature Stack*

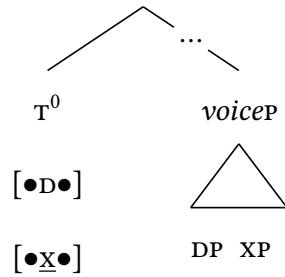
$$[\underline{\text{TNS}}] > [\bullet\text{ADV}\bullet] > [\bullet\text{BO}\bullet] > [\bullet\text{D}\bullet] > [\underline{\text{T}}] > [*\phi*] > [\underline{\text{C}}] > [\bullet\text{WH}\bullet]$$

4.3 Reprojection and Suspension

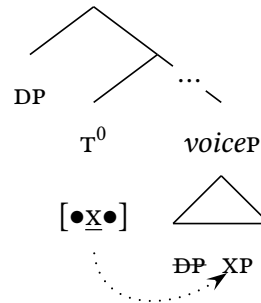
We can now draw the central series of links from *WH*-movement to structure-building. To derive a *v2* system where only subject-initial clauses project to the level of the TP , we can posit two ordered features on a bundled head τ^0 (77a). The first of these is $[\bullet\text{D}\bullet]$, which should always attract the closest *DP* and thus draw the subject to SPEC,TP . The second feature must then be $[\bullet\underline{\text{x}}\bullet]$: one that attracts topics, foci, and *WH*-phrases and carries an additional trigger for unfolding (which must be able to attach to derivational features like $[\bullet\underline{\text{x}}\bullet]$ in the same way that it attaches to categorial features like $[\underline{\text{C}}]$, as a *SECOND-ORDER* feature (Adger & Svenonius, 2011)). Once the derivation reaches $[\bullet\underline{\text{x}}\bullet]$, this feature triggers a second round of probing beneath τ^0 (77b). In the absence of a licit target for movement, it must then fail (in the sense of Preminger 2011): when it cannot force attraction, the derivation simply moves on to the remaining features on τ^0 . If this head carries no further features that drive reprojection, unfolding will stop—and if no further heads are externally merged above the TP , the ensuing clause will be a TP (77c).

(77) *Reprojection for Attraction*

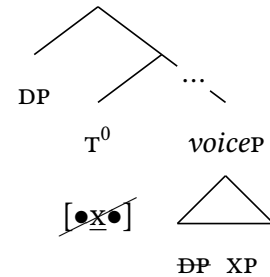
a. *Two Features on T^0*



b. *Probing for $[\bullet \underline{X} \bullet]$*



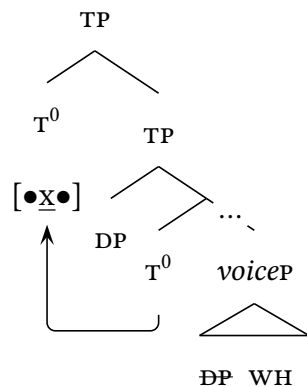
c. *Deactivating $[\bullet \underline{X} \bullet]$*



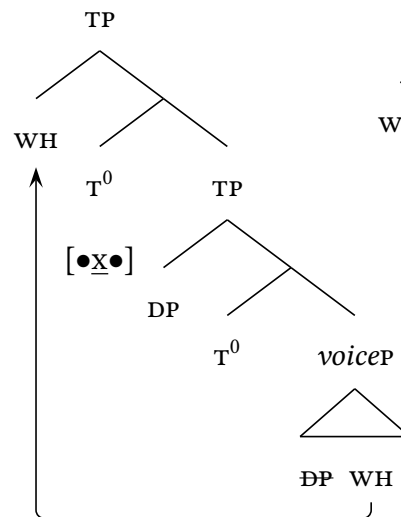
In the derivations that feature a lower WH-phrase, however, $[\bullet \underline{X} \bullet]$ forces the step of unfolding in (78a): one that allows $[\bullet \underline{X} \bullet]$ to excorporate, undergo head-movement, and reproject to attract the WH to the higher position in (78b). I will assume that this step constructs a higher instance of the category TP, as it is driven by $[\bullet \underline{X} \bullet]$ and not a category feature like $[\underline{c}]$. If it forces this higher position to host the typical exponent of T^0 , on a par with the regular process of T^0 -to- C^0 , it must then be followed by the step of morphological head-movement in (78c): one that operates in the postsyntax, after Harizanov & Gribanova 2019, in response to another second-order feature on $[\bullet \underline{X} \bullet]$.

(78) *Reprojection for Attraction*

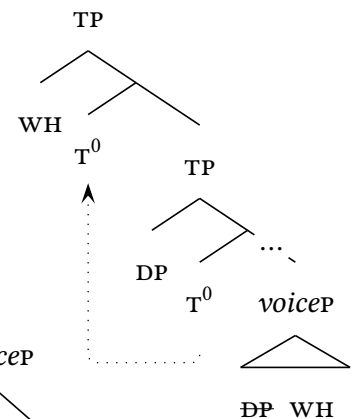
a. *Reprojection*



b. *Attraction*



c. *Amalgamation*



We can now turn to interactions of the target type: those where structure-building is suspended by the A-movement of WH-subjects. We can begin by deriving the analysis of English in Rizzi 1997: one on which regular embedded clauses host the heads FORCE^0 and FIN^0 (79a) but the clauses that launch subject extraction require the bundling in (79b).

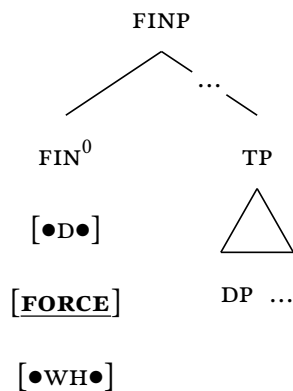
(79) *Subject Extraction $\rightarrow$ Less Structure*

- a. REGULAR CLAUSE: $[\text{FORCE} \quad \text{that}_{\text{FORCE}^0} \quad [\text{FIN} \quad \emptyset_{\text{FIN}^0} \quad [\text{TP} \quad \text{DP}_{\text{SUBJ}} \quad \dots]]]$
- b. SUBJECT EXTRACTION: $[\text{FORCE/FIN} \quad \text{---}_{\text{WH}} \quad \emptyset_{\text{FORCE}^0/\text{FIN}^0} \quad [\text{TP} \quad \text{---}_{\text{WH}} \quad \dots]]]$

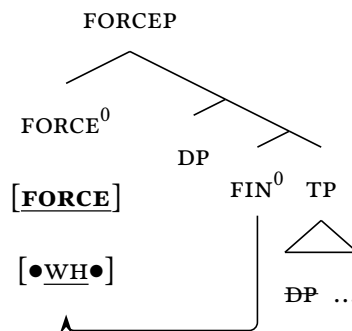
To capture this interaction, we can appeal to a derivational pressure for local economy (for an alternative that derives similar effects through transderivational comparison, see Martinović 2015, 2022). I propose that FIN^0 and FORCE^0 typically enter the derivation as a bundled head with three features in English: $[\bullet\text{D}\bullet]$, $[\bullet\text{WH}\bullet]$, and an intermediate category feature $[\text{FORCE}]$ (80a). This bundle typically unfolds to yield the syntax in (80b): the subject is drawn to SPEC,FINP by a step of A-movement and $[\text{FORCE}]$ reprojects to build the FORCEP . This step is unlinked to the feature $[\bullet\text{WH}\bullet]$, and as a result, it occurs without regard to the presence of a lower WH-phrase. If $[\text{FORCE}]$ lacks the lexical property that triggers amalgamation with FIN^0 in the postsyntax, in turn, this step should allow the separate exponence of FIN^0 and FORCE^0 , yielding the emergence of *that* (80c).

(80) *Reprojection for Structure-Building*

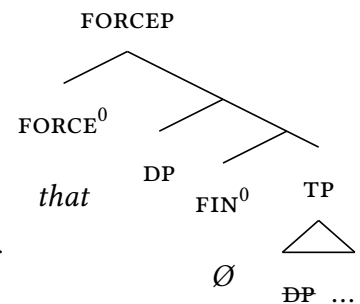
a. *The Bundled c^0*



b. *Reprojection*



c. *Exponence*



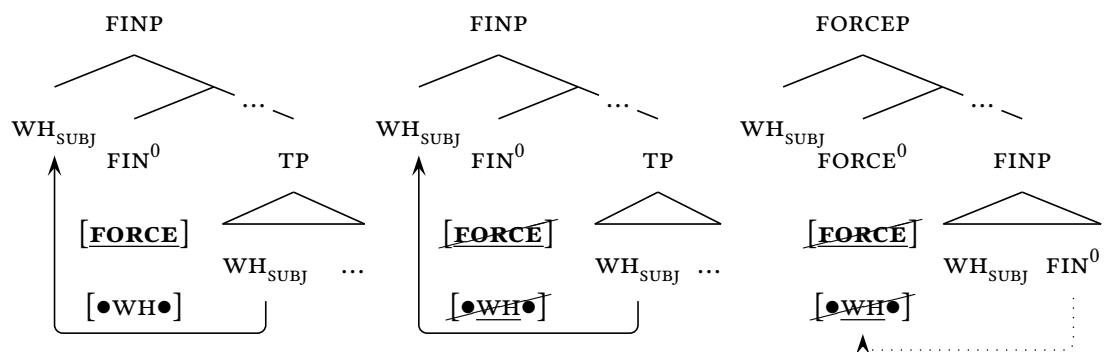
In the embedded clauses that host WH-subjects, however, unfolding must be derailed. The breakdown begins from a pattern of visibility: while the features on bundled heads are typically checked in a rigid order, I propose that they are all visible to the derivation when the bundled head sits at the root (mirroring the early Minimalist view that all features on a head can be accessed at once: Chomsky 1995; Longenbaugh 2019). In this configuration, we can force an interaction through the FREE RIDER CONDITION of Chomsky 1995: a pressure that forces a head to check as many features as possible once it begins to interact with a particular goal (see also Bruening 2001; Rezac 2013). When $[\bullet D \bullet]$ attracts a WH-subject to SPEC,FINP, more specifically, the derivation must recognize that it can leverage the same goal to check the feature $[\bullet WH \bullet]$, which is typically checked only after **[FORCE]** (81a). I thus propose that the system of economy forces the parasitic feature-checking in (81b): the derivation skips **[FORCE]** to immediately check $[\bullet WH \bullet]$. In the ensuing configuration, it is impossible to build a FORCEP (81c).

(81) *Subject Extraction: Feature-Skipping*

a. *The Starting Point*

b. *Feature Skipping*

c. ~~X~~ *Reprojection*

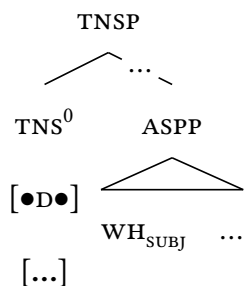


I propose that the same interaction delivers the Mandar pattern of bundling to TNSP. The possibility for feature-skipping emerges in the clauses where absolutive WH-phrases raise to SPEC,ASPP (82a), where they are poised to satisfy the feature $[\bullet D \bullet]$ on the bundled head TNS/T/C⁰ (82a). When they check this second feature through A-movement to SPEC,TNSP, the derivation identifies the possibility for the step of multitasking in (82b): one that parasitically checks the later feature $[\bullet WH \bullet]$ by skipping all of the features that

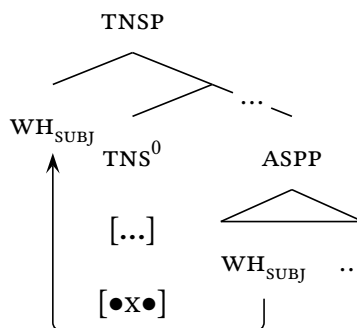
canonically separate it from [$\bullet D \bullet$]. In this configuration, the Free Rider Condition demands the same result: the derivation must skip through the intervening features that build the TP, trigger absolutive agreement, and ultimately construct the CP (82c). As a consequence, the clauses that host WH-subjects will never project beyond the TNSP.

(82) *Subject Extraction: Deep Suspension*

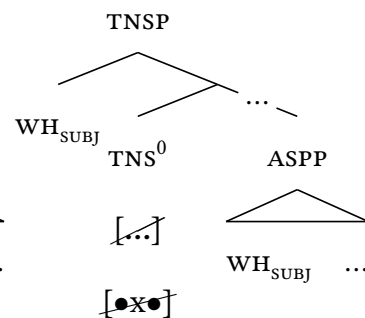
a. *The Starting Point*



b. *Step One: A-Mvmt*



c. *Step Two: Skipping*



The result is a framework that delivers these patterns of reduction in a flexible and theoretically integrated way: with the right type of bundling and distribution of features, the A-movement of WH-phrases can derail the construction of the canonical TP and CP. The pattern of bundling that we have posited for English triggers a fairly minor disruption at the level of the CP, preserving subject agreement in τ^0 but yielding a comp-trace effect that turns on the bundling of c^0 . The more extensive pattern that we have posited in Mandar, in turn, delivers a comp-trace effect and a type of Anti-Agreement that involves bundling deeper into the TP. The differences between these systems emerge squarely in the lexicon, where universally-ordered category features are bundled together, interspersed with derivational features, and selected as triggers for reprojection in a language-particular way. Within the derivation, however, these bundled heads are unfolded within a single system of feature-based economy: they undergo a cyclic process of head-movement and reprojection, driven by the distribution of reprojection features, that can be derailed when the triggers for A-movement attract targets that satisfy features from the $\bar{A}$ -domain.

5 Conclusion

Stepping back from the guys of the system, we can now take stock on the conceptual shifts that follow from the analytical and theoretical steps that we have taken up to this point. At the most basic level, our investigation has established three truths about Anti-Agreement: (i) some Anti-Agreement Effects do emerge from the special syntax of subject extraction, (ii) these effects need not implicate direct extraction from positions in the extended VP, and (iii) in the systems where WH-subjects undergo A-movement through the middle field, Anti-Agreement can emerge from a total syntactic suppression of agreeing heads like τ^0 . These results run against the narrow thrust of [Baier 2018](#), who attempts to consign all Anti-Agreement to the morphology, and they flesh out the wider typology of these effects in a useful way: establishing that Anti-Agreement can emerge along this specific derivational path and showing that it can still be linked to the syntax of subject extraction in the ergative systems where the highest argument is the absolutive DP.

Turning to the wider syntax of subject extraction, we can then lay out a series of results around the movement of the absolutive DP. Starting in the system of quantifier stranding, the selectional profile and linear distribution of the WH-quantifier *pira* establish a new argument that WH-movement can strand quantifiers along its path. The facts of stranding, in turn, reveal that the clauses which descriptively launch subject extraction require WH-subjects to raise through a regular subject position that falls just below the suppressed τ^0 . The transparency effects in the systems of raising and hyperraising suggest that structure-building is suspended just as WH-subjects raise to this position, yielding an argument that the Anti-Agreement and comp-trace effects in this system involve the non-construction of the TP and CP. These results, at last, suggest that WH-subjects are positioned through A-movement to a regular subject position, rather than $\bar{A}$ -movement to SPEC,CP—providing us with the tools to explain why subject extraction in Mandar takes such a different shape.

Returning to the empirical terrain, at last, this theoretical step allows us to understand a final effect that we have set aside up to this point. When WH-subjects undergo long-distance extraction, they are usually carried through the higher clauses along their paths by a regular type of $\bar{A}$ -movement that proceeds through SPEC,CP, rather than the special process that targets WH-subjects alone (Pesetsky, 2019; Bošković, 2024). In the systems that show Anti-Agreement, this tendency is then matched by the following fact: subject agreement is typically only derailed in the lowest clauses along the path (Ouhalla, 1993; Ouali, 2008). In Mandarin, however, this is not so: when embedded WH-subjects undergo long-distance extraction, they trigger Anti-Agreement in every clause on the way (83).

(83) *Long-Distance Wh-Dependencies: Successive-Cyclic Anti-Agreement*

- a. Innai mu-irrangngi ___/(*i) [_{MODP} marondong lamba ___]?
 who 2ERG-hear 3ABS tomorrow go
 ‘Who did you hear is leaving tomorrow?’
- b. **Innai** mu-irrangngi ___/(*i) [_{MODP} na-sa’ding ___/(*i) [_{MODP} na-pile ___]]?
 who 2ERG-hear 3ABS 3ERG-feel 3ABS 3ERG-pick
 ‘Who did you hear he feels they’ll pick?’

This restriction then correlates in the usual way with the systematic suppression of intermediate complementizers (84a) and the emergence of long-distance jailbreak effects (84b). These patterns, in turn, are matched by the usual realignment in the system of Condition C to suggest that all of these clauses fail to raise and only project to the MODP.

(84) *Long-Distance Wh-Dependencies: Comp-Trace Effects + Jailbreaks*

- a. Innai na-pissangngi [_{MODP} (***mua**) na-irrangngi [_{MODP} marondong lamba]]?
 who 3ERG-announce that 3ERG-hear tomorrow go
 ‘Who did they announce they heard would go tomorrow?’
- b. Innai mu-irrangngi **bo** [_{MODP} na-sa’ding iCicci’ [_{MODP} na-tumae ___]]
 who 2ERG-hear again 3ERG-feel NAME 3ERG-propose
 ‘Who did you hear that Chichi’ feels will propose to her again?’

In the intermediate clauses along this path, moreover, the facts of *pira*-stranding reveal that the moved WH-phrase must pass through every typical position of the absolutive DP: for instance, every intermediate SPEC,VP (85a) and every intermediate SPEC,ASPP (85b).

(85) *Long-Distance Wh-Dependencies: Raising through Absolutive Positions*

- a. Innai mu-irrangngi [_{MODP} na-sa'ding iKaco' __ **pira** dionging [_{MODP} pole]]?
 who 2ERG-hear 3ERG-feel NAME all yesterday come
 'Who all did you hear that Kacho' felt yesterday would come?'
- b. Innai mu-irrangngi [_{MODP} dionging __ **pira** na-sa'ding iKaco' [_{MODP} pole]]?
 who 2ERG-hear yesterday all 3ERG-feel NAME come
 'Who all did you hear that yesterday Kacho' felt would come?'

These facts suggest that WH-subjects are carried to the edges of their clauses of origin by a process that feeds raising through all the canonical A-positions of the absolutive DP. The step that positions them in the embedded SPEC,MODP, more specifically, must feed raising into a higher SPEC,VP—and from there, SPEC,voiceP, SPEC,ASPP, and SPEC,MODP. The result is that cases of long-distance subject extraction in Mandar cannot involve an initial step of subject extraction and then successive-cyclic $\bar{A}$ -movements through intermediate SPEC,CPS: rather, they must involve continuous A-movement up to the matrix SPEC,MODPP.

At a language-internal level, this final observation allows us to reinforce and extend the claim that WH-subjects are carried to their final landing sites by the A-syntax alone. In comparative perspective, however, it is significant for the way that it fits into a regional generalization: many Western Austronesian languages seem to require long-distance WH-subject extraction to implicate raising as well (Finer 2003; Keenan 2008; Gerassimova & Sells 2008; Davies & Kurniawan 2013). It may thus be that in those language, as in Mandar, subject WH-phrases are positioned by an A-movement that halts construction of the CP.

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